



Qualification Pack

Machine Operator Tool Room

QP Code: CPC/CAP/Q0601

Version: 1.0

NSQF Level: 3.5

CENTRAL INSTITUTE OF PETROCHEMICALS ENGINEERING & TECHNOLOGY || CIPET Head Office, T.V.K.
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Qualification Pack

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CPC/CAP/Q0601: Machine Operator Tool Room

Brief Job Description

Machine Operator - Tool Room qualification is to get acquainted with setting up and operating the Tool Room machine to produce good quality products from raw materials.

Personal Attributes

Technical aptitude, Safety-conscious, Reliability and punctuality, Organizational skills, Problem-solving ability, Teamwork and communication, Quality-focused mindset, Willingness to learn

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CPC/CAP/N0401: Basic health and safety practices at the workplace, 5S](#)
2. [CPC/CAP/N0601: To Understanding the basic concepts, design, drawings and planning for machining components, making tools & dies and coordinating with others](#)
3. [CPC/CAP/N0602: To Perform fitting operations on machining components using hand tools](#)
4. [CPC/CAP/N0603: To Operation on Drilling](#)
5. [CPC/CAP/N0604: To Perform Operation on Shaping Machine](#)
6. [CPC/CAP/N0605: Operation on Lathe machine](#)
7. [CPC/CAP/N0606: To Perform Operation on Milling Machines](#)
8. [CPC/CAP/N0607: To Perform Operation on Grinding machines](#)
9. [CPC/CAP/N0608: To Preform Basic programming and operation on CNC Machines](#)
10. [CPC/CAP/N0403: Effective working with others.](#)
11. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Chemical & PetroChemical
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Sub-Sector	Petrochemicals
Occupation	Tool Room Plastics, Tool Room Plastics
Country	India
NSQF Level	3.5
Credits	20
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7223
Minimum Educational Qualification & Experience	11th Class OR 10th Class (or Equivalent) with 1.5 years of experience Relevant Experience OR 8th Class with 4.5 years of experience Relevant Experience OR Previous relevant Qualification of NSQF Level (3) with 1.5 years of experience Relevant Experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	25/05/2028
NSQC Approval Date	26/05/2025
Version	1.0
Reference code on NQR	QG-3.5-CP-04124-2025-V2-CIPET
NQR Version	2

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CPC/CAP/N0401: Basic health and safety practices at the workplace, 5S

Description

This unit is about creating a Safe and Healthy work place, adhering to the safety guidelines in the working area, following practices which are not impacting the environment in a negative manner. This NOS is about ensuring all 5S activities both at the shop floor and the office area to facilitate increase in work productivity. It covers knowledge of fire safety, common first aid applications, safe practice and emergency applications.

Scope

The scope covers the following :

- The role holder will be responsible for
- Identifying and reporting of risks
- Creating and sustaining a safe, clean and environment friendly Work place.
- Ensure sorting, streamlining & organizing, storage and documentation, cleaning, standardization and sustenance across the plant and office premises of the organization

Elements and Performance Criteria

Health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** Wear protective clothing/equipment for specific tasks and work conditions
- PC2.** Carry out safe working practices while dealing with hazards to ensure the safety of Self and others.
- PC3.** Apply good housekeeping practices at all times

Fire safety

To be competent, the user/individual on the job must be able to:

- PC4.** Use the various appropriate fire extinguishers on different types of fires correctly
- PC5.** Demonstrate rescue techniques applied during fire hazard, demonstrate good housekeeping in order to prevent fire hazards, demonstrate the correct use of a fire extinguisher.

Identify and report the risks identified

To be competent, the user/individual on the job must be able to:

- PC6.** Identify activities which can cause potential injury through sharp objects, burns, fall, electricity, gas leakages, radiation, poisonous fumes, chemicals, loud noise, and Identify areas in the plant which are potentially hazardous/ unhygienic in nature. Conduct regular checks with support of the maintenance team on machine health to identify potential hazards due to wear and tear of machine.
- PC7.** Inform the concerned authorities on the potential risks identified in the processes, workplace area/ layout, materials used etc, Inform the concerned authorities about machine breakdowns, damages which can potentially harm man/ machine during operations.
- PC8.** Create awareness amongst other by sharing information on the identified risks.

Ensure sorting

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To be competent, the user/individual on the job must be able to:

- PC9.** Follow the sorting process and check that the tools, fixtures & jigs that are lying on workstations are the ones in use and unnecessary items are not cluttering the workbenches or work surfaces.
- PC10.** Ensure segregation of waste in hazardous/ non Hazardous waste as per the sorting work instructions
- PC11.** Follow the technique of waste disposal and waste storage in the proper bins as per SOP
- PC12.** Segregate the items which are labeled as red tag items for the process area and keep them in the correct places
- PC13.** Sort the tools/ equipment/ fasteners/ spare parts as per specifications/ utility into proper trays, cabinets, lockers as mentioned in the 5S guidelines/ work instructions
- PC14.** Ensure that areas of material storage areas are not overflowing
- PC15.** Properly stack the various types of boxes and containers as per the size/ utility to avoid any fall of items/ breakage and also enable easy sorting when required
- PC16.** Return the extra material and tools to the designated sections and make sure that no additional material/ tool is lying near the work area
- PC17.** Follow the floor markings/ area markings used for demarcating the various sections in the plant as per the prescribed instructions and standards

Ensure proper documentation and storage(organizing, streamlining)

To be competent, the user/individual on the job must be able to:

- PC18.** Follow the proper labelling mechanism of instruments/ boxes/ containers and maintaining reference files/ documents with the codes and the lists
- PC19.** Check that the items in the respective areas have been identified as broken or damaged
- PC20.** Follow the given instructions and check for labelling of fluids, oils, lubricants, solvents, chemicals etc. and proper storage of the same. To avoid spillage, leakage, fire etc.
- PC21.** Make sure that all material and tools are stored in the designated places and in the manner indicated in the 5S instructions

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Relevant standards, procedures and policies related to Health, Safety and Environment followed in the company
- KU2.** Emergency handling procedures & hierarchy for escalation
- KU3.** Basic knowledge of Safety procedures(firefighting, first aid) within the organization
- KU4.** Basic knowledge of various types of PPEs and their usage
- KU5.** Basic knowledge of risks/hazards associated with each occupation in the organization
- KU6.** Knowledge of personal hygiene and how an individual can contribute towards creating a highly safe and clean working environment
- KU7.** Meaning of hazards and risks
- KU8.** Health and safety hazards commonly present in the work environment and related precautions

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- KU9.** Possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible
- KU10.** Possible causes of risk and accident (due to oil leakage)
- KU11.** Safe working practices when working with tools and machines
- KU12.** Safe working practices while working at various hazardous sites
- KU13.** Where to find all the general health and safety equipment in the workplace
- KU14.** Various dangers associated with the use of electrical equipment
- KU15.** Preventative and remedial actions to be taken in the case of exposure to toxic materials
- KU16.** Importance of using protective clothing/equipment while working
- KU17.** Precautionary activities to prevent the fire accident
- KU18.** Various causes of fire
- KU19.** Techniques of using the different fire extinguishers
- KU20.** Different methods of extinguishing fire
- KU21.** Different materials used for extinguishing fire
- KU22.** Rescue techniques applied during a fire hazard
- KU23.** Various types of safety signs and what they mean
- KU24.** Appropriate basic first aid treatment relevant to the condition eg. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries
- KU25.** Content of written accident report
- KU26.** Potential injuries and ill health associated with incorrect manual handling
- KU27.** Safe lifting and carrying practices
- KU28.** Personal safety, health and dignity issues relating to the movement of a person by others
- KU29.** Potential impact to a person who is moved incorrectly
- KU30.** Have basic knowledge of 5S procedures
- KU31.** Know various types 5s practices followed in various areas
- KU32.** Understand the 5S checklists provided in the department/ team
- KU33.** Have skills to identify useful & non useful items
- KU34.** Have knowledge of labels , signs & colors used as indicators
- KU35.** Have knowledge on how to sort and store various types of tools, equipment, material etc.
- KU36.** Know , how to identify various types of waste products
- KU37.** Understand the impact of waste/ dirt/ dust/unwanted substances on the process/ environment/ machinery/ human body.
- KU38.** Have knowledge of best ways of cleaning & waste disposal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Write basic level notes and observations
- GS2.** Read safety instructions put up across the plant premises

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- GS3.** Read safety precautions mentioned in equipment manuals and panels to understand the potential risks associated
- GS4.** Effectively communicate information to team members
- GS5.** Inform employees in the plant and concerned functions about events, incidents & potential risks observed related to Safety, Health and Environment.
- GS6.** Question operator/ supervisor in order to understand the safety related issues
- GS7.** Attentively listen with full attention and comprehend the information given by the speaker during safety drills and training programs
- GS8.** Plan and organize the work order and jobs received from the internal customers
- GS9.** Plan and organize the design documents received from internal customers
- GS10.** Organize all process/ equipment manuals so that sorting out information is Fast
- GS11.** Use common sense and make judgments during day to day basis
- GS12.** Use reasoning skills to identify and resolve basic problems
- GS13.** Use intuition to detect any potential problems which could arise during operations
- GS14.** Follow instructions and work on areas of improvement identified
- GS15.** Complete the assigned tasks with minimum supervision
- GS16.** Complete the job defined by the supervisor within the timelines and quality norms

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Health and safety</i>	1.5	6	-	-
PC1. Wear protective clothing/equipment for specific tasks and work conditions	0.5	2	-	-
PC2. Carry out safe working practices while dealing with hazards to ensure the safety of Self and others.	0.5	2	-	-
PC3. Apply good housekeeping practices at all times	0.5	2	-	-
<i>Fire safety</i>	1	4	-	-
PC4. Use the various appropriate fire extinguishers on different types of fires correctly	0.5	2	-	-
PC5. Demonstrate rescue techniques applied during fire hazard, demonstrate good housekeeping in order to prevent fire hazards, demonstrate the correct use of a fire extinguisher.	0.5	2	-	-
<i>Identify and report the risks identified</i>	1.5	6	-	-
PC6. Identify activities which can cause potential injury through sharp objects, burns, fall, electricity, gas leakages, radiation, poisonous fumes, chemicals, loud noise, and Identify areas in the plant which are potentially hazardous/ unhygienic in nature. Conduct regular checks with support of the maintenance team on machine health to identify potential hazards due to wear and tear of machine.	0.5	2	-	-
PC7. Inform the concerned authorities on the potential risks identified in the processes, workplace area/ layout, materials used etc, Inform the concerned authorities about machine breakdowns, damages which can potentially harm man/ machine during operations.	0.5	2	-	-
PC8. Create awareness amongst other by sharing information on the identified risks.	0.5	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Ensure sorting</i>	4	10	-	-
PC9. Follow the sorting process and check that the tools, fixtures & jigs that are lying on workstations are the ones in use and unnecessary items are not cluttering the workbenches or work surfaces.	0.5	2	-	-
PC10. Ensure segregation of waste in hazardous/ non Hazardous waste as per the sorting work instructions	0.5	2	-	-
PC11. Follow the technique of waste disposal and waste storage in the proper bins as per SOP	0.5	1	-	-
PC12. Segregate the items which are labeled as red tag items for the process area and keep them in the correct places	0.5	1	-	-
PC13. Sort the tools/ equipment/ fasteners/ spare parts as per specifications/ utility into proper trays, cabinets, lockers as mentioned in the 5S guidelines/ work instructions	0.5	1	-	-
PC14. Ensure that areas of material storage areas are not overflowing	0.25	0.5	-	-
PC15. Properly stack the various types of boxes and containers as per the size/ utility to avoid any fall of items/ breakage and also enable easy sorting when required	0.25	0.5	-	-
PC16. Return the extra material and tools to the designated sections and make sure that no additional material/ tool is lying near the work area	0.5	1	-	-
PC17. Follow the floor markings/ area markings used for demarcating the various sections in the plant as per the prescribed instructions and standards	0.5	1	-	-
<i>Ensure proper documentation and storage(organizing, streamlining)</i>	2	4	-	-
PC18. Follow the proper labelling mechanism of instruments/ boxes/ containers and maintaining reference files/ documents with the codes and the lists	0.5	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC19. Check that the items in the respective areas have been identified as broken or damaged	0.5	1	-	-
PC20. Follow the given instructions and check for labelling of fluids, oils, lubricants, solvents, chemicals etc. and proper storage of the same. To avoid spillage, leakage, fire etc.	0.5	1	-	-
PC21. Make sure that all material and tools are stored in the designated places and in the manner indicated in the 5S instructions	0.5	1	-	-
NOS Total	10	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0401
NOS Name	Basic health and safety practices at the workplace, 5S
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Generic
NSQF Level	3.5
Credits	1
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

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CPC/CAP/N0601: To Understanding the basic concepts, design, drawings and planning for machining components, making tools & dies and coordinating with others

Description

This NOS covers the planning and co-ordination for machining components, manufacturing of tools and die as per given dimension. The candidate should understand the design and drawings, identifying the sequence of operations required by studying the design and co-ordination with other for the performance of required operations on the metal component. The candidate will further get the work done from the various operators and workers. The candidate will be expected to execute the job unsupervised, taking responsibility as a part of the team for the quality and accuracy of the work produced

Scope

The scope covers the following :

- This unit/task covers the following:
- Working safely
- Understanding design requirements and planning
- Co-coordinating with others

Elements and Performance Criteria

Working safely

To be competent, the user/individual on the job must be able to:

- PC1.** • adhere to procedures or systems in place for health and safety, personal
• protective equipment (PPE) and other relevant safety regulations
- PC2.** ensure all the machines used are in a safe and useable condition
- PC3.** ensure that all the tool room machines are correctly guarded at all times

Understanding design requirements and planning

To be competent, the user/individual on the job must be able to:

- PC4.** obtain sample component/ drawings and other engineering information as per company procedures
- PC5.** identify requirements by analyzing sample component, design and drawing
- PC6.** Plan sequence of operations for machining component keeping in mind various considerations like requirements, timelines, resources available, interdependencies, constraints, compliances, etc
- PC7.** report and rectify cases of inappropriate information in design documents as per organizational procedures
- PC8.** compute dimensions, sizes, shapes and tolerances of machining component are as per specifications and as per company procedures
- PC9.** determine information such as number of parts to make, engineered components and material to be used, and machines to be used

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- PC10.** • identify and confirm resources required such as components, machinery, range of materials and processes
• Range of Materials: Ferrous metals: eg. carbon steels, stainless steels, cast iron, tool steel, hard metals; Non-ferrous metals: eg. bronze, bronze alloys, copper and copper alloys
- PC11.** identify the operations that will be required for machining components based on design requirements
- PC12.** identify type of equipment required for machining components based on the operations selected
- PC13.** estimate timelines for each task accurately
- PC14.** establish work completion time by determining a schedule of operations
- PC15.** obtain necessary approvals for the action plan
- PC16.** allocate responsibilities to machine operators as per the operations selected
- PC17.** ensure that the machine operators are clear about the sequence of activities, priorities and considerations

Co-ordinating with others

To be competent, the user/individual on the job must be able to:

- PC18.** release drawings and machining specifications to machine operators
- PC19.** identify and select machines for machining components based on design and drawings
- PC20.** identify and select cutting tools based on design and drawings
- PC21.** select and procure appropriate metals to be used for machining components as per design requirement
- PC22.** hand over cutting tools and raw material for machining to the machine operators
- PC23.** handle all clarifications sought by the operators
- PC24.** collect job from all operators
- PC25.** check the jobs as per drawing/instruction
- PC26.** ensure in-process inspection of the machining component

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** policies and procedures followed in the company relevant to own employment and performance conditions
- KU2.** health and safety requirements in the work place
- KU3.** working in clean and safe environment
- KU4.** Job responsibilities and information pertaining to employment terms, entitlements, job role and responsibilities
- KU5.** • reporting mechanism, department functions and procedures in the
• work place
- KU6.** Related workforce and their responsibilities within the work area
- KU7.** procedures for reporting at work and employment related issues
- KU8.** documentation and related procedures applicable related to employment and work

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- KU9.** Documentation in context of employment and work
- KU10.** The sources for information about job specifications
- KU11.** read and prepare various types of job specification documents for job requirements
- KU12.** hazards associated with the activities
- KU13.** the various fitting activities to be carried out
- KU14.** how to extract and use information from engineering drawings and related specifications in relation to work undertaken
- KU15.** various hand fitting methods
- KU16.** how to interpret first and third angle drawings
- KU17.** conventional machining methods
- KU18.** basic knowledge of accessing computer drawing using Auto-Cad software to be used for viewing designs drawings
- KU19.**
 - factors that affect the selection of cutting speed, feed and depth of cut
 - Factors: type of material, size of material, operations being performed, work holding devices, condition of machine, finish required, tolerances
- KU20.** the British and metric systems of measurement
- KU21.** geometric dimensioning and tolerance
- KU22.** how to plan and organize the team
- KU23.** machine operations and sequencing
- KU24.** machine capacity and capabilities
- KU25.** types of machine tools such as lathes, milling, drills, grinding
- KU26.** work holding devices and equipment
- KU27.** machining accessories
- KU28.** limits and capabilities of tooling, accessories and holding devices
- KU29.** How to check the work piece and the measuring instruments
- KU30.** need to check that the component using calibrated measuring instrument
- KU31.** properties of metals
- KU32.** heat treatment processes of tool steel

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret information correctly from various job specification documents, manuals, health and safety instructions, etc
- GS2.** fill up appropriate technical forms, process charts, log sheet as per organizational format in English and/or local language
- GS3.** convey and share technical information clearly using appropriate language
- GS4.** check and clarify task-related information
- GS5.** liaise with appropriate authorities using correct protocol
- GS6.** communicate with people in respectful form and manner in line with organizational protocol

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- GS7.**
 - Understanding mathematical calculations and geometry
 - arithmetic: addition, subtraction, multiplication, division, fractions and decimals, percentages and proportions, simple ratios and averages
- GS8.** use appropriate measuring techniques
- GS9.** Understanding mathematical calculations to a degree of accuracy that is appropriate to the value being calculated
- GS10.** use a calculator to raise a number to a power and determine square roots
- GS11.** calculation of the value of angles in a triangle
- GS12.** maintain current knowledge of process developments
- GS13.** clarify job related information with appropriate personnel or technical adviser
- GS14.** seek to improve and modify own work practices
- GS15.** perform basic operations in a computer like switching it on/off, using the mouse and keyboard, accessing files, opening, closing, creating and deleting folders, etc
- GS16.** use basic office applications like spread sheet, word processor, presentations
- GS17.** use email to communicate within the organization as per organization guidelines
- GS18.** retrieve and enter data using standard system forms and templates
- GS19.** take printouts of documents
- GS20.** identify problems with work planning, procedures, output and behavior and their implications
- GS21.** prioritize and plan for problem solving
- GS22.** communicate problems appropriately to others
- GS23.** identify sources of information and support for problem solving
- GS24.** seek assistance and support from other sources to solve problems
- GS25.** identify effective resolution methods
- GS26.** select and apply resolution techniques
- GS27.** seek evidence for problem resolution
- GS28.** plan, prioritize and sequence work operations as per job requirements
- GS29.** organize and analyze information relevant to work
- GS30.** basic concepts of shop-floor work productivity including minimize rejection and optimization of time
- GS31.** undertake and express new ideas and initiatives to others
- GS32.** modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
- GS33.** one competencies in new and different situations and contexts to achieve more
- GS34.** exercise restraint while expressing dissent and during conflict situations
- GS35.** avoid and manage distractions to be disciplined at work
- GS36.** manage own time for achieving better results
- GS37.** work in a team in order to achieve better results
- GS38.** identify and clarify work roles within a team
- GS39.** communicate and cooperate with others in the team for better results
- GS40.** seek assistance from fellow team members

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Working safely</i>	0.75	3	-	-
PC1. • adhere to procedures or systems in place for health and safety, personal • protective equipment (PPE) and other relevant safety regulations	0.25	1	-	-
PC2. ensure all the machines used are in a safe and useable condition	0.25	1	-	-
PC3. ensure that all the tool room machines are correctly guarded at all times	0.25	1	-	-
<i>Understanding design requirements and planning</i>	6	10.5	-	-
PC4. obtain sample component/ drawings and other engineering information as per company procedures	0.25	1	-	-
PC5. identify requirements by analyzing sample component, design and drawing	0.25	1	-	-
PC6. Plan sequence of operations for machining component keeping in mind various considerations like requirements, timelines, resources available, interdependencies, constraints, compliances, etc	0.25	1	-	-
PC7. report and rectify cases of inappropriate information in design documents as per organizational procedures	0.25	1	-	-
PC8. compute dimensions, sizes, shapes and tolerances of machining component are as per specifications and as per company procedures	0.5	1	-	-
PC9. determine information such as number of parts to make, engineered components and material to be used, and machines to be used	0.5	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. - identify and confirm resources required such as components, machinery, range of materials and processes - Range of Materials: Ferrous metals: eg. carbon steels, stainless steels, cast iron, tool steel, hard metals; Non-ferrous metals: eg. bronze, bronze alloys, copper and copper alloys	0.5	1	-	-
PC11. identify the operations that will be required for machining components based on design requirements	0.5	0.5	-	-
PC12. identify type of equipment required for machining components based on the operations selected	0.5	0.5	-	-
PC13. estimate timelines for each task accurately	0.5	0.5	-	-
PC14. establish work completion time by determining a schedule of operations	0.5	0.5	-	-
PC15. obtain necessary approvals for the action plan	0.5	0.5	-	-
PC16. allocate responsibilities to machine operators as per the operations selected	0.5	0.5	-	-
PC17. ensure that the machine operators are clear about the sequence of activities, priorities and considerations	0.5	0.5	-	-
<i>Co-ordinating with others</i>	3.25	6.5	-	-
PC18. release drawings and machining specifications to machine operators	0.5	0.5	-	-
PC19. identify and select machines for machining components based on design and drawings	0.5	0.5	-	-
PC20. identify and select cutting tools based on design and drawings	0.5	0.5	-	-
PC21. select and procure appropriate metals to be used for machining components as per design requirement	0.5	0.5	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. hand over cutting tools and raw material for machining to the machine operators	0.25	0.5	-	-
PC23. handle all clarifications sought by the operators	0.25	1	-	-
PC24. collect job from all operators	0.25	1	-	-
PC25. check the jobs as per drawing/instruction	0.25	1	-	-
PC26. ensure in-process inspection of the machining component	0.25	1	-	-
NOS Total	10	20	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0601
NOS Name	To Understanding the basic concepts, design, drawings and planning for machining components, making tools & dies and coordinating with others
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Tool Room Plastics, Tool Room Plastics
NSQF Level	3.5
Credits	2
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQF Clearance Date	26/05/2025

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CPC/CAP/N0602: To Perform fitting operations on machining components using hand tools

Description

This unit covers fitting of machining components using hand tools and manually operated machines, to form the shape of a component from raw material, as per given specifications in the drawing. This involves carrying out the fitting operations like filing, drilling, chiseling, threading, tapping, scraping, manual lapping and shaping in order to fit a component as per specifications. The candidate will be expected to perform under minimum supervision, taking self-interest at work and for the quality and accuracy of the work

Scope

The scope covers the following :

- This unit/task covers the following:
- Working safely
- Preparing for fitting operations
- Marking components
- Performing fitting operation

Elements and Performance Criteria

Working safely

To be competent, the user/individual on the job must be able to:

- PC1.** comply with health and safety, environmental and other relevant regulations and guidelines at work
- PC2.** adhere to procedures and guidelines for personal protective equipment (PPE) and other relevant safety regulations while performing die fitting operations
- PC3.** work following laid down procedures and instructions
- PC4.** ensure work area is clean and safe from hazards
- PC5.** ensure that all tools, equipment, power tool cables, extension leads are in a safe and usable condition

Preparing for fitting operations

To be competent, the user/individual on the job must be able to:

- PC6.**
 - obtain job specification from a valid and approved source
 - Valid sources: job instruction sheet/job card; work drawings and instructions; planning documentation; quality control documents; operation sheets; instructions from supervisor
- PC7.**
 - read and understand job requirements from the job specification document properly
 - Job specification documents: detailed component drawings; approved detailed drawings; reference tables and charts
 - Job requirements: raw materials required (type, quality, quantity); dimensions; limit, fit and tolerances; operations required; shape to be machined; instruments and tools to be used; interdependencies; timelines of completion

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- PC8.**
 - report and rectify incorrect information in job specification
 - documents as per job requirement
- PC9.** preparation for the fitting operations as per procedure
- PC10.** ensure that all calibrated measuring instruments used
- PC11.** ensure that the components used are free from foreign objects, dirt and corrosion
- PC12.** obtain correct work pieces and consumables as per job requirements
- PC13.** obtain appropriate tools and measuring instruments
- PC14.** Setting of work pieces as per job requirements using appropriate holding devices

Marking components

To be competent, the user/individual on the job must be able to:

- PC15.** Marking specified features with the help of marking-out methods on the work pieces as per job specification by using appropriate measuring and marking tools
- PC16.** mark out templates for tracing/transferring the specified features on the work pieces as per drawing
- PC17.** Tracing or transfer the specified features from the templates onto the work pieces as per drawing

Performing fitting operations on machining components using hand tools

To be competent, the user/individual on the job must be able to:

- PC18.** perform fitting operations on various forms of metal components using a range of hand tools and manually operated machines
- PC19.** follow the specified machining sequence and procedure as per job specifications
- PC20.** check the machined components to ensure completeness of work
- PC21.** check the quality of the output as per required standards, using visual checks and measurement of dimensional parameters using measuring instruments
- PC22.** produce components with various features as per standards applicable to the process
- PC23.** Check the finished components as per job requirement
- PC24.** complete documentation during and post operations as per procedures
- PC25.** return all tools and equipment to the correct location on completion of the fitting activities
- PC26.** leave the work area in a safe and tidy condition on completion of job activities

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** policies and procedures followed in the company relevant to own employment and performance conditions
- KU2.** health and safety requirements in the work place
- KU3.** working in clean and safe environment
- KU4.** Job responsibilities and information related to employment terms, entitlements, job role and responsibilities
- KU5.**
 - reporting mechanism, department functions and procedures in the
 - work place

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- KU6.** Related workforce and their responsibilities within the work area
- KU7.** procedures for reporting at work and employment related issues
- KU8.** documentation and related procedures applicable related to employment and work
- KU9.** Documentation in connection with employment and work
- KU10.** specific safe working practices, fitting procedures
- KU11.** hazards associated with carrying out the fitting operations and how can they be minimized
- KU12.** personal protective equipment to be used during the fitting activities and where can it be obtained
- KU13.** types and sources of appropriate job specifications
- KU14.** common terminology used in fitting
- KU15.** importance of following specified fitting sequences and procedures
- KU16.** importance and procedures of ensuring suitability of work piece and consumables for the specified job
- KU17.** tools and equipment used for the fitting operations
- KU18.** importance and procedures to ensure that tools and equipment are in a safe and usable condition
- KU19.** correct techniques and procedures to carry out specific fitting operations by hand tools and manually operated machines
- KU20.** importance of securing the work piece correctly using appropriate devices and mechanisms
- KU21.** common problems that can occur in the fitting operations and their implications
- KU22.** correct procedures to address problems commonly encountered during fitting operations
- KU23.** importance of reporting problems immediately and accurately
- KU24.** meaning and importance of quality in relation to final and intermediate job output
- KU25.** how to check the correctness of the shaped components against the specified quality standards
- KU26.** range of materials used in relevant fitting applications
- KU27.** relevant mechanical properties of metals and implications for job
- KU28.** importance of using correct procedures as per type and form of materials and metal components

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and interpret information correctly from various job specification documents, manuals, health and safety instructions, etc
- GS2.** fill up appropriate technical forms, process charts, log sheet as per organizational format
- GS3.** convey and share technical information clearly using appropriate language
- GS4.** check and clarify task-related information
- GS5.** liaise with appropriate authorities using correct protocol
- GS6.** communicate with people in respectful form and manner in line with organizational protocol

Qualification Pack

- GS7.** undertake numerical operations, and calculations/ formulae
- GS8.** identify and draw various basic, compound and solid shapes as per dimensions given
- GS9.** use appropriate measuring techniques and units of measurement
- GS10.** use appropriate units and number systems to express degree of accuracy
- GS11.** interpret and express tolerance in terms of limits on dimensions
- GS12.** calculation of the value of angles in a triangle
- GS13.** maintain current knowledge of applicable standards, codes of practice and product/process developments
- GS14.** participate in on-the-job and other learning, training and development interventions and assessment
- GS15.** clarify task related information with appropriate personnel or technical adviser
- GS16.** seek to improve and modify own work practices
- GS17.** identify problems with work planning, procedures, output and behavior and their implications
- GS18.** prioritize and plan for problem solving
- GS19.** communicate problems appropriately to others
- GS20.** identify sources of information and support for problem solving
- GS21.** seek assistance and support from other sources to solve problems
- GS22.** identify effective resolution techniques
- GS23.** select and apply resolution techniques
- GS24.** seek evidence for problem resolution
- GS25.** plan, prioritize and sequence work operations as per job requirements
- GS26.** organize and analyze information relevant to work
- GS27.** basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time
- GS28.** undertake and express new ideas and initiatives to others
- GS29.** modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
- GS30.** participate in improvement procedures including process, quality and customer relationships
- GS31.** competencies in new and different situations to achieve more
- GS32.** exercise restraint while expressing dissent and during conflict situations
- GS33.** avoid and manage distractions to be disciplined at work
- GS34.** manage the time for achieving better results
- GS35.** work in a team in order to achieve better results
- GS36.** identify and clarify work roles within a team
- GS37.** communicate and cooperate with others in the team for better results
- GS38.** seek assistance from fellow team members

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Working safely</i>	2.5	5	-	-
PC1. comply with health and safety, environmental and other relevant regulations and guidelines at work	0.5	1	-	-
PC2. adhere to procedures and guidelines for personal protective equipment (PPE) and other relevant safety regulations while performing die fitting operations	0.5	1	-	-
PC3. work following laid down procedures and instructions	0.5	1	-	-
PC4. ensure work area is clean and safe from hazards	0.5	1	-	-
PC5. ensure that all tools, equipment, power tool cables, extension leads are in a safe and usable condition	0.5	1	-	-
<i>Preparing for fitting operations</i>	4.5	9	-	-
PC6. • obtain job specification from a valid and approved source • Valid sources: job instruction sheet/job card; work drawings and instructions; planning documentation; quality control documents; operation sheets; instructions from supervisor	0.5	1	-	-
PC7. • read and understand job requirements from the job specification document properly • Job specification documents: detailed component drawings; approved detailed drawings; reference tables and charts • Job requirements: raw materials required (type, quality, quantity); dimensions; limit, fit and tolerances; operations required; shape to be machined; instruments and tools to be used; interdependencies; timelines of completion	0.5	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC8. • report and rectify incorrect information in job specification • documents as per job requirement	0.5	1	-	-
PC9. preparation for the fitting operations as per procedure	0.5	1	-	-
PC10. ensure that all calibrated measuring instruments used	0.5	1	-	-
PC11. ensure that the components used are free from foreign objects, dirt and corrosion	0.5	1	-	-
PC12. obtain correct work pieces and consumables as per job requirements	0.5	1	-	-
PC13. obtain appropriate tools and measuring instruments	0.5	1	-	-
PC14. Setting of work pieces as per job requirements using appropriate holding devices	0.5	1	-	-
<i>Marking components</i>	2.5	3	-	-
PC15. Marking specified features with the help of marking-out methods on the work pieces as per job specification by using appropriate measuring and marking tools	0.5	1	-	-
PC16. mark out templates for tracing/transferring the specified features on the work pieces as per drawing	1	1	-	-
PC17. Tracing or transfer the specified features from the templates onto the work pieces as per drawing	1	1	-	-
<i>Performing fitting operations on machining components using hand tools</i>	5.5	8	-	-
PC18. perform fitting operations on various forms of metal components using a range of hand tools and manually operated machines	1	1	-	-
PC19. follow the specified machining sequence and procedure as per job specifications	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC20. check the machined components to ensure completeness of work	0.5	0.5	-	-
PC21. check the quality of the output as per required standards, using visual checks and measurement of dimensional parameters using measuring instruments	0.5	0.5	-	-
PC22. produce components with various features as per standards applicable to the process	0.5	1	-	-
PC23. Check the finished components as per job requirement	0.5	1	-	-
PC24. complete documentation during and post operations as per procedures	0.5	1	-	-
PC25. return all tools and equipment to the correct location on completion of the fitting activities	0.5	1	-	-
PC26. leave the work area in a safe and tidy condition on completion of job activities	0.5	1	-	-
NOS Total	15	25	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0602
NOS Name	To Perform fitting operations on machining components using hand tools
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Tool Room Plastics, Tool Room Plastics
NSQF Level	3.5
Credits	2
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

CPC/CAP/N0603: To Operation on Drilling

Description

This NOS is covers performing drilling operations on drilling machine to produce a range of ferrous and Nonferrous metals that combine a number of different features e.g. drilled, bored and reamed holes. The candidate will be expected to work as per instructions given in design/drawing, taking personal responsibility for own actions and for the quality and accuracy of the work. The candidate will follow the safety instructions required when working with the Drilling machine, its associated cutting tools and equipment. The candidate will be required to demonstrate safe working practices in tool room, and will understand the responsibility they owe to themselves and others in the workplace.

Scope

The scope covers the following :

- This unit/task covers the following:
- Working safely
- Preparing for operating Drilling machine
- Carrying out operations on Drilling machine

Elements and Performance Criteria

Working safely

To be competent, the user/individual on the job must be able to:

- PC1.** comply with health and safety, environmental and other relevant regulations and guidelines at work
- PC2.** adhere to procedures and guidelines for personal protective equipment (PPE) and other relevant safety regulations while performing machining operations
- PC3.** ensure work area is clean and safe from hazards
- PC4.** ensure that all tools, equipment, power tool cables, extension leads are in a safe and usable condition
- PC5.** ensure that machine guards are in place and are correctly fitted
- PC6.** read and understand safety instructions, warning signs on the machine

Preparing for operating conventional turning machine

To be competent, the user/individual on the job must be able to:

- PC7.** ensure that all measuring equipment is within calibration due date
- PC8.** ensure availability of job specification from a valid source/Drawings
- PC9.**
 - read and establish job requirements from the job specification document Job specifications documents: detailed component drawings; approved
 - drawings /illustrations; national, international and organizational standards
- PC10.** ensure that the incoming components used are free from foreign objects, dirt or other contamination
- PC11.** prepare and maintain the work area as per procedure or operation specification

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- PC12.** plan to carry out the required drilling activities and the sequence of operations as per specifications
- PC13.** apply safe working practices and procedures at all times
- PC14.** obtain all the appropriate materials, cutting tools and measuring equipments required for the drilling operations
- PC15.** confirm that the machine is ready for production
- PC16.** prepare for the Drilling activities by mounting, positioning and correctly setting a range of work holding devices and cutting tools
- PC17.** seek any necessary instruction/training on the operation of the machine, where required
- PC18.** hold components securely, without distortion
- PC19.** ensure that machine settings are adjusted as and when required to maintain the required accuracy

Carrying out operations on conventional turning machine

To be competent, the user/individual on the job must be able to:

- PC20.** obtain the component drawings, specifications, job instructions required for the components to be machined
- PC21.** use and extract information from engineering drawings and related specifications (to include symbols and conventions to appropriate IS or ISO standards in relation to work undertaken)
- PC22.** set and adjust the machine tool speeds and feeds to achieve the component specification
- PC23.** mount and set the required work holding devices, work piece and cutting tools
- PC24.** operate the machine tool controls safely and correctly, in line with operational procedures
- PC25.** control the machine in both hand and power modes for normal operations
- PC26.** stop the machine in both normal and emergency situations correctly, and follow right procedure for restarting after an emergency
- PC27.** use drilling machine accessories that consists of vices, drill chuck, sleeves, clamps, tool holders
- PC28.** position and secure work holding devices to the machine spindle
- PC29.** perform drilling operations using various equipments to produce components with various features
- PC30.** produce components as per given quality standards
- PC31.** plan and work to achieve given production targets
- PC32.** overcome the effects of backlash in machine slides and screws
- PC33.** perform the technique of trial cut for checking dimensional accuracy
- PC34.** apply roughing and finishing cuts, considering the effect on tool life, surface finish and dimensional accuracy
- PC35.** use cutting fluids for different materials
- PC36.** use range of measuring instruments to check critical parameters
- PC37.** clamp the work piece in a chuck/work holding device
- PC38.** perform the checks to be carried out on the components before removing them from the machine, and the equipment needed for this activity
- PC39.** ensure that the quality control procedures are used while operating the equipment

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** relevant health and safety requirements applicable in the work place
- KU3.** importance of working in clean and safe environment
- KU4.** own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
- KU5.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU6.** relevant people and their responsibilities within the work area
- KU7.** procedures for reporting work and employment related issues
- KU8.** documentation and related procedures applicable in the context of employment and work
- KU9.** importance and purpose of documentation in context of employment and work
- KU10.** where personal protective equipment to be worn can be obtained
- KU11.** where to obtain the component drawings, specifications and/or job instructions required for them components to be machined KB3. hazards associated with the drilling operations and how they can be minimized
- KU12.** meaning and purpose of drilling
- KU13.** safety mechanisms on the machine, and the procedure for checking that they function correctly
- KU14.** how to tighten all the bolts and other securing devices securely
- KU15.** importance of keeping the work area clean and tidy
- KU16.** how to use metric systems of measurement
- KU17.** main features of the Drilling machine and the accessories that can be used

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret information correctly from various job specification documents, manuals, health and safety instructions, etc. applicable to the job
- GS2.** fill up appropriate technical forms, process charts, activity logs as per organizational format
- GS3.** fill up appropriate technical forms, process charts, activity logs as per organizational format
- GS4.** check and clarify task-related information
- GS5.** liaise with appropriate authorities using correct protocol
- GS6.** communicate with people in respectful form and manner in line with organizational protocol
- GS7.** undertake numerical operations, and calculations/ formulae
- GS8.** identify and draw various basic, compound and solid shapes as per dimensions given
- GS9.** use appropriate measuring techniques and units of measurement
- GS10.** use appropriate units and number systems to express degree of accuracy

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- GS11.** use metric systems of measurement
- GS12.**
 - participate in on-the-job and other learning, training and development
 - interventions and assessments
- GS13.** clarify task related information with appropriate personnel or technical adviser
- GS14.** seek to improve and modify own work practices
- GS15.** maintain current knowledge of application standards, codes of practice and product/process developments
- GS16.** identify problems with work planning, procedures, output and behavior and their implications
- GS17.** prioritize and plan for problem solving
- GS18.** communicate problems appropriately to others
- GS19.** identify sources of information and support for problem solving
- GS20.** seek assistance and support from other sources to solve problems
- GS21.** identify effective resolution techniques
- GS22.** select and apply resolution techniques
- GS23.** seek evidence for problem resolution
- GS24.** plan, prioritize and sequence work operations as per job requirements
- GS25.** organize and analyze information relevant to work
- GS26.** basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time
- GS27.** undertake and express new ideas and initiatives to others
- GS28.** modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
- GS29.** participate in improvement procedures including process, quality and internal/external customer/supplier relationships
- GS30.** one competencies in new and different situations and contexts to achieve more
- GS31.** exercise restraint while expressing dissent and during conflict situations
- GS32.** avoid and manage distractions to be disciplined at work
- GS33.** manage own time for achieving better results
- GS34.** work in a team in order to achieve better results
- GS35.** identify and clarify work roles within a team
- GS36.** communicate and cooperate with others in the team for better results
- GS37.** seek assistance from fellow team members

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Working safely</i>	3	6	-	-
PC1. comply with health and safety, environmental and other relevant regulations and guidelines at work	0.5	1	-	-
PC2. adhere to procedures and guidelines for personal protective equipment (PPE) and other relevant safety regulations while performing machining operations	0.5	1	-	-
PC3. ensure work area is clean and safe from hazards	0.5	1	-	-
PC4. ensure that all tools, equipment, power tool cables, extension leads are in a safe and usable condition	0.5	1	-	-
PC5. ensure that machine guards are in place and are correctly fitted	0.5	1	-	-
PC6. read and understand safety instructions, warning signs on the machine	0.5	1	-	-
<i>Preparing for operating conventional turning machine</i>	6.5	13	-	-
PC7. ensure that all measuring equipment is within calibration due date	0.5	1	-	-
PC8. ensure availability of job specification from a valid source/Drawings	0.5	1	-	-
PC9. - read and establish job requirements from the job specification document Job specifications documents: detailed component drawings; approved - drawings /illustrations; national, international and organizational standards	0.5	1	-	-
PC10. ensure that the incoming components used are free from foreign objects, dirt or other contamination	0.5	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. prepare and maintain the work area as per procedure or operation specification	0.5	1	-	-
PC12. plan to carry out the required drilling activities and the sequence of operations as per specifications	0.5	1	-	-
PC13. apply safe working practices and procedures at all times	0.5	1	-	-
PC14. obtain all the appropriate materials, cutting tools and measuring equipments required for the drilling operations	0.5	1	-	-
PC15. confirm that the machine is ready for production	0.5	1	-	-
PC16. prepare for the Drilling activities by mounting, positioning and correctly setting a range of work holding devices and cutting tools	0.5	1	-	-
PC17. seek any necessary instruction/training on the operation of the machine, where required	0.5	1	-	-
PC18. hold components securely, without distortion	0.5	1	-	-
PC19. ensure that machine settings are adjusted as and when required to maintain the required accuracy	0.5	1	-	-
<i>Carrying out operations on conventional turning machine</i>	10.5	21	-	-
PC20. obtain the component drawings, specifications, job instructions required for the components to be machined	0.5	1	-	-
PC21. use and extract information from engineering drawings and related specifications (to include symbols and conventions to appropriate IS or ISO standards in relation to work undertaken)	0.5	1	-	-
PC22. set and adjust the machine tool speeds and feeds to achieve the component specification	0.5	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. mount and set the required work holding devices, work piece and cutting tools	0.5	1	-	-
PC24. operate the machine tool controls safely and correctly, in line with operational procedures	0.5	1	-	-
PC25. control the machine in both hand and power modes for normal operations	0.5	1	-	-
PC26. stop the machine in both normal and emergency situations correctly, and follow right procedure for restarting after an emergency	0.5	1	-	-
PC27. use drilling machine accessories that consists of vices, drill chuck, sleeves, clamps, tool holders	0.5	1	-	-
PC28. position and secure work holding devices to the machine spindle	0.5	1	-	-
PC29. perform drilling operations using various equipments to produce components with various features	0.5	1	-	-
PC30. produce components as per given quality standards	0.5	1	-	-
PC31. plan and work to achieve given production targets	0.5	1	-	-
PC32. overcome the effects of backlash in machine slides and screws	0.5	1	-	-
PC33. perform the technique of trial cut for checking dimensional accuracy	0.5	1	-	-
PC34. apply roughing and finishing cuts, considering the effect on tool life, surface finish and dimensional accuracy	0.5	1	-	-
PC35. use cutting fluids for different materials	0.5	1	-	-
PC36. use range of measuring instruments to check critical parameters	0.5	1	-	-
PC37. clamp the work piece in a chuck/work holding device	0.5	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC38. perform the checks to be carried out on the components before removing them from the machine, and the equipment needed for this activity	0.5	1.5	-	-
PC39. ensure that the quality control procedures are used while operating the equipment	1	1.5	-	-
NOS Total	20	40	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0603
NOS Name	To Operation on Drilling
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Tool Room Plastics, Tool Room Plastics
NSQF Level	3.5
Credits	2
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

CPC/CAP/N0604: To Perform Operation on Shaping Machine

Description

This NOS is covers performing shaping operations on shaping machine to produce a range of ferrous and Nonferrous metals that combine a number of different features e.g. parallel, stepped, angular machining. The candidate will be expected to work as per instructions given in design/drawing, taking personal responsibility for own actions and for the quality and accuracy of the work. The candidate will follow the safety instructions required when working with the shaping machine, its associated cutting tools and equipment. The candidate will be required to demonstrate safe working practices in tool room, and will understand the responsibility they owe to themselves and others in the workplace.

Scope

The scope covers the following :

- This unit/task covers the following:
- Working safely
- Preparing for operating shaping machine
- Carrying out operations on shaping machine

Elements and Performance Criteria

Working safely

To be competent, the user/individual on the job must be able to:

- PC1.** comply with health and safety, environmental and other relevant regulations and guidelines at work
- PC2.** adhere to procedures and guidelines for personal protective equipment (PPE) and other relevant safety regulations while performing machining operations
- PC3.** ensure work area is clean and safe from hazards
- PC4.** ensure that all tools, equipment, power tool cables, extension leads are in a safe and usable condition
- PC5.** ensure that machine guards are in place and are correctly fitted.
- PC6.** read and understand safety instructions, warning signs on the machine

Preparing for operating conventional turning machine

To be competent, the user/individual on the job must be able to:

- PC7.** ensure that all measuring equipment is within calibration due date
- PC8.** ensure availability of job specification from a valid source/Drawings
- PC9.** read and establish job requirements from the job specification document
- PC10.** ensure that the incoming components used are free from foreign objects, dirt or other contamination
- PC11.** prepare and maintain the work area as per procedure or operation specification
- PC12.** plan to carry out the required shaping activities and the sequence of operations as per specifications

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- PC13.** apply safe working practices and procedures at all times
- PC14.** obtain all the appropriate materials, cutting tools and measuring equipments required for the shaping operations
- PC15.** confirm that the machine is ready for production
- PC16.** prepare for the shaping activities by mounting, positioning and correctly setting a range of work holding devices and cutting tools
- PC17.** seek any necessary instruction/training on the operation of the machine, where required
- PC18.** hold components securely, without distortion
- PC19.** hold components securely, without distortion

Carrying out operations on conventional turning machine

To be competent, the user/individual on the job must be able to:

- PC20.** obtain the component drawings, specifications, job instructions required for the components to be machined
- PC21.** use and extract information from engineering drawings and related specifications (to include symbols and conventions to appropriate IS or ISO standards in relation to work undertaken)
- PC22.** set and adjust the machine tool speeds and feeds to achieve the component specification
- PC23.** mount and set the required work holding devices, work piece and cutting tools
- PC24.** operate the machine tool controls safely and correctly, in line with operational procedures
- PC25.** control the machine in both hand and power modes for normal operations
- PC26.** stop the machine in both normal and emergency situations correctly, and follow right procedure for restarting after an emergency
- PC27.** use shaping machine accessories that consists of vices, clamps, tool holders
- PC28.** position and secure work holding devices to the machine ram
- PC29.** perform shaping operations using various equipments to produce components with various features
- PC30.** produce components as per given quality standards
- PC31.** plan and work to achieve given production targets
- PC32.** overcome the effects of backlash in machine slides
- PC33.** perform the technique of trial cut for checking dimensional accuracy
- PC34.** apply roughing and finishing cuts, considering the effect on tool life, surface finish and dimensional accuracy
- PC35.** use cutting fluids for different materials
- PC36.** use range of measuring instruments to check critical parameters
- PC37.** clamp the work piece in a chuck/work holding device
- PC38.** perform the checks to be carried out on the components before removing them from the machine, and the equipment needed for this activity
- PC39.** ensure that the quality control procedures are used while operating the equipment

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** relevant health and safety requirements applicable in the work place
- KU3.** importance of working in clean and safe environment
- KU4.** own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
- KU5.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU6.** relevant people and their responsibilities within the work area
- KU7.** procedures for reporting work and employment related issues
- KU8.** documentation and related procedures applicable in the context of employment and work
- KU9.** importance and purpose of documentation in context of employment and work
- KU10.** where personal protective equipment to be worn can be obtained
- KU11.** where to obtain the component drawings, specifications and/or job instructions required for them components to be machined
- KU12.** hazards associated with the shaping operations and how they can be minimized
- KU13.** meaning and purpose of shaping
- KU14.** safety mechanisms on the machine, and the procedure for checking that they function correctly
- KU15.** how to tighten all the bolts and other securing devices securely
- KU16.** importance of keeping the work area clean and tidy
- KU17.** how to use metric systems of measurement
- KU18.** main features of the shaper and the accessories that can be used
- KU19.** classification and purpose of various accessories
- KU20.** tool materials (classification, properties and use)
- KU21.** how to identify the factors that affect the selection of cutting feeds and speeds, and the depth of cut that can be taken
- KU22.** the shaping operations that can be performed using various equipment, and the component features produced on metal and non-metal components
- KU23.** effects of backlash in machine slides and screws, and how this can be overcome
- KU24.** safety instructions and warning signs on the machine
- KU25.** types of cutting fluids and their properties
- KU26.** effects of clamping the work piece in a work holding device, and how this can cause distortion in the finished components
- KU27.** problems that can occur with the shaping and how these can be overcome
- KU28.** correct equipment and procedure to use for checking critical quality parameters
- KU29.** production cost, machine hour rate, raw material cost, tool cost, coolant cost, overheads, cycle time, idle time, cost of machine idling, part rejection cost
- KU30.** selection of cutting tools, tool materials, selecting cutting parameters from tool catalogues, selecting coolant
- KU31.** relationship between surface finish, tool angle, speed and feed rate
- KU32.** impact of depth of cut on chatter, surface finish

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KU33. extent of their own authority and to whom they should report if they have problems that they cannot resolve

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret information correctly from various job specification documents, manuals, health and safety instructions, etc. applicable to the job
- GS2.** fill up appropriate technical forms, process charts, activity logs as per organizational format
- GS3.** convey and share technical information clearly using appropriate language
- GS4.** check and clarify task-related information
- GS5.** liaise with appropriate authorities using correct protocol
- GS6.** communicate with people in respectful form and manner in line with organizational protocol
- GS7.** undertake numerical operations, and calculations/ formulae
- GS8.** identify and draw various basic, compound and solid shapes as per dimensions given
- GS9.** use appropriate measuring techniques and units of measurement
- GS10.** use appropriate units and number systems to express degree of accuracy
- GS11.** use metric systems of measurement
- GS12.**
 - participate in on-the-job and other learning, training and development
 - interventions and assessments
- GS13.** clarify task related information with appropriate personnel or technical adviser
- GS14.** seek to improve and modify own work practices
- GS15.** maintain current knowledge of application standards, codes of practice and product/process developments
- GS16.** identify problems with work planning, procedures, output and behavior and their implications
- GS17.** prioritize and plan for problem solving
- GS18.** communicate problems appropriately to others
- GS19.** identify sources of information and support for problem solving
- GS20.** seek assistance and support from other sources to solve problems
- GS21.** identify effective resolution techniques
- GS22.** select and apply resolution techniques
- GS23.** seek evidence for problem resolution
- GS24.** plan, prioritize and sequence work operations as per job requirements
- GS25.** organize and analyze information relevant to work
- GS26.** basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time
- GS27.** undertake and express new ideas and initiatives to others
- GS28.** modify work plan to overcome unforeseen difficulties or developments that occur as work progresses



Qualification Pack

- GS29.** participate in improvement procedures including process, quality and internal/external customer/supplier relationships
- GS30.** one competencies in new and different situations and contexts to achieve more
- GS31.** exercise restraint while expressing dissent and during conflict situations
- GS32.** avoid and manage distractions to be disciplined at work
- GS33.** manage own time for achieving better results
- GS34.** work in a team in order to achieve better results
- GS35.** identify and clarify work roles within a team
- GS36.** communicate and cooperate with others in the team for better results
- GS37.** seek assistance from fellow team members

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Working safely</i>	3	6	-	-
PC1. comply with health and safety, environmental and other relevant regulations and guidelines at work	0.5	1	-	-
PC2. adhere to procedures and guidelines for personal protective equipment (PPE) and other relevant safety regulations while performing machining operations	0.5	1	-	-
PC3. ensure work area is clean and safe from hazards	0.5	1	-	-
PC4. ensure that all tools, equipment, power tool cables, extension leads are in a safe and usable condition	0.5	1	-	-
PC5. ensure that machine guards are in place and are correctly fitted.	0.5	1	-	-
PC6. read and understand safety instructions, warning signs on the machine	0.5	1	-	-
<i>Preparing for operating conventional turning machine</i>	6.5	13	-	-
PC7. ensure that all measuring equipment is within calibration due date	0.5	1	-	-
PC8. ensure availability of job specification from a valid source/Drawings	0.5	1	-	-
PC9. read and establish job requirements from the job specification document	0.5	1	-	-
PC10. ensure that the incoming components used are free from foreign objects, dirt or other contamination	0.5	1	-	-
PC11. prepare and maintain the work area as per procedure or operation specification	0.5	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. plan to carry out the required shaping activities and the sequence of operations as per specifications	0.5	1	-	-
PC13. apply safe working practices and procedures at all times	0.5	1	-	-
PC14. obtain all the appropriate materials, cutting tools and measuring equipments required for the shaping operations	0.5	1	-	-
PC15. confirm that the machine is ready for production	0.5	1	-	-
PC16. prepare for the shaping activities by mounting, positioning and correctly setting a range of work holding devices and cutting tools	0.5	1	-	-
PC17. seek any necessary instruction/training on the operation of the machine, where required	0.5	1	-	-
PC18. hold components securely, without distortion	0.5	1	-	-
PC19. hold components securely, without distortion	0.5	1	-	-
<i>Carrying out operations on conventional turning machine</i>	10.5	21	-	-
PC20. obtain the component drawings, specifications, job instructions required for the components to be machined	0.5	1	-	-
PC21. use and extract information from engineering drawings and related specifications (to include symbols and conventions to appropriate IS or ISO standards in relation to work undertaken)	0.5	1	-	-
PC22. set and adjust the machine tool speeds and feeds to achieve the component specification	0.5	1	-	-
PC23. mount and set the required work holding devices, work piece and cutting tools	0.5	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. operate the machine tool controls safely and correctly, in line with operational procedures	0.5	1	-	-
PC25. control the machine in both hand and power modes for normal operations	0.5	1	-	-
PC26. stop the machine in both normal and emergency situations correctly, and follow right procedure for restarting after an emergency	0.5	1	-	-
PC27. use shaping machine accessories that consists of vices, clamps, tool holders	0.5	1	-	-
PC28. position and secure work holding devices to the machine ram	0.5	1	-	-
PC29. perform shaping operations using various equipments to produce components with various features	0.5	1	-	-
PC30. produce components as per given quality standards	0.5	1	-	-
PC31. plan and work to achieve given production targets	0.5	1	-	-
PC32. overcome the effects of backlash in machine slides	0.5	1	-	-
PC33. perform the technique of trial cut for checking dimensional accuracy	0.5	1	-	-
PC34. apply roughing and finishing cuts, considering the effect on tool life, surface finish and dimensional accuracy	0.5	1	-	-
PC35. use cutting fluids for different materials	0.5	1	-	-
PC36. use range of measuring instruments to check critical parameters	0.5	1	-	-
PC37. clamp the work piece in a chuck/work holding device	0.5	1.5	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC38. perform the checks to be carried out on the components before removing them from the machine, and the equipment needed for this activity	0.5	1.5	-	-
PC39. ensure that the quality control procedures are used while operating the equipment	1	1	-	-
NOS Total	20	40	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0604
NOS Name	To Perform Operation on Shaping Machine
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Tool Room Plastics, Tool Room Plastics
NSQF Level	3.5
Credits	1
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

CPC/CAP/N0605: Operation on Lathe machine

Description

This NOS is covers performing turning operations on Lathe machine such as centre lathes to produce a range of ferrous and Non ferrous metals that combine a number of different features e.g. parallel, stepped and taper turning, machining diameters, drilled, bored and reamed holes, internal, external threading operation and turning special form profiles. The candidate will be expected to work as per instructions given in design/drawing, taking personal responsibility for own actions and for the quality and accuracy of the work. The candidate will follow the safety instructions required when working with the Lathe machine, its associated cutting tools and equipment. The candidate will be required to demonstrate safe working practices in tool room, and will understand the responsibility they owe to themselves and others in the workplace

Scope

The scope covers the following :

- This unit/task covers the following:
- Working safely
- Preparing for operating conventional turning machine
- Carrying out operations on conventional turning machine

Elements and Performance Criteria

Working safely

To be competent, the user/individual on the job must be able to:

- PC1.** comply with health and safety, environmental and other relevant regulations and guidelines at work
- PC2.** adhere to procedures and guidelines for personal protective equipment (PPE) and other relevant safety regulations while performing turning operations
- PC3.** ensure work area is clean and safe from hazards
- PC4.** ensure that all tools, equipment, power tool cables, extension leads are in a safe and usable condition
- PC5.** ensure that machine guards are in place and are correctly fitted
- PC6.** read and understand safety instructions, warning signs on the machine

Preparing for operating conventional turning machine

To be competent, the user/individual on the job must be able to:

- PC7.** ensure that all measuring equipment is within calibration due date
- PC8.** ensure availability of job specification from a valid source/Drawings
- PC9.** read and establish job requirements from the job specification document
- PC10.** ensure that the incoming components used are free from foreign objects, dirt or other contamination
- PC11.** prepare and maintain the work area as per procedure or operation specification

Qualification Pack

- PC12.** plan to carry out the required turning activities and the sequence of operations as per specifications
- PC13.** apply safe working practices and procedures at all times
- PC14.** obtain all the appropriate materials, cutting tools and measuring equipments required for the turning operation
- PC15.** confirm that the machine is ready for production
- PC16.** prepare for the turning activities by mounting, positioning and correctly setting a range of work holding devices and cutting tools
- PC17.** seek any necessary instruction/training on the operation of the machine, where required
- PC18.** hold components securely, without distortion
- PC19.** ensure that machine settings are adjusted as and when required to maintain the required accuracy

Carrying out operations on conventional turning machine

To be competent, the user/individual on the job must be able to:

- PC20.** obtain the component drawings, specifications, job instructions required for the components to be machined
- PC21.** use and extract information from engineering drawings and related specifications
- PC22.** set and adjust the machine tool speeds and feeds to achieve the component specification
- PC23.** mount and set the required work holding devices, work piece and cutting tools
- PC24.** operate the machine tool controls safely and correctly, in line with operational procedures
- PC25.** control the machine in both hand and power modes for normal operations
- PC26.** Stop the machine in both normal and emergency situations correctly, and follow right procedure for restarting after an emergency
- PC27.** use lathes and the accessories that consists of saddle, capstan/turret head, compound slide, tailstock, taper turning attachments, profile attachments, fixed and travelling steadies
- PC28.** position and secure work holding devices to the machine spindle
- PC29.** perform turning operations using various equipments to produce components with various features
- PC30.** produce components as per given quality standards
- PC31.** plan and work to achieve given production targets
- PC32.** overcome the effects of backlash in machine slides and screws
- PC33.** perform the technique of trial cut for checking dimensional accuracy
- PC34.** apply roughing and finishing cuts, considering the effect on tool life, surface finish and dimensional accuracy
- PC35.** use cutting fluids for different materials
- PC36.** use range of measuring instruments to check critical parameters
- PC37.** clamp the work piece in a chuck/work holding device
- PC38.** perform the checks to be carried out on the components before removing them from the machine, and the equipment needed for this activity
- PC39.** ensure that the quality control procedures are used while operating the equipment

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** relevant health and safety requirements applicable in the work place
- KU3.** importance of working in clean and safe environment
- KU4.** own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
- KU5.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU6.** relevant people and their responsibilities within the work area
- KU7.** procedures for reporting work and employment related issues
- KU8.** documentation and related procedures applicable in the context of employment and work
- KU9.** importance and purpose of documentation in context of employment and work
- KU10.** where personal protective equipment to be worn can be obtained
- KU11.** where to obtain the component drawings, specifications and/or job instructions required for them components to be machined
- KU12.** hazards associated with the turning operations and how they can be minimized
- KU13.** meaning and purpose of turning
- KU14.** safety mechanisms on the machine, and the procedure for checking that they function correctly
- KU15.** how to tighten all the bolts, cam locks or other securing devices securely
- KU16.** importance of keeping the work area clean and tidy
- KU17.** how to use metric systems of measurement
- KU18.** main features of the lathes and the accessories that can be used
- KU19.** classification and purpose of various accessories
- KU20.** tool materials (classification, properties and use)
- KU21.** how to identify the factors that affect the selection of cutting feeds and speeds, and the depth of cut that can be taken
- KU22.** the Turning operations that can be performed using various equipment, and the component features produced on metal and non-metal components
- KU23.** effects of backlash in machine slides and screws, and how this can be overcome
- KU24.** safety instructions and warning signs on the machine
- KU25.** types of cutting fluids and their properties
- KU26.** effects of clamping the work piece in a chuck/work holding device, and how this can cause distortion in the finished components
- KU27.** problems that can occur with the turning activities, and how these can be overcome
- KU28.** correct equipment and procedure to use for checking critical quality parameters
- KU29.** production cost, machine hour rate, raw material cost, tool cost, coolant cost, overheads, cycle time, idle time, cost of machine idling, part rejection cost
- KU30.** selection of cutting tools, tool materials, chip breaker geometry, selecting

Generic Skills (GS)

Qualification Pack

User/individual on the job needs to know how to:

- GS1.** read and interpret information correctly from various job specification documents, manuals, health and safety instructions, etc. applicable to the job
- GS2.** fill up appropriate technical forms, process charts, activity logs as per organizational format
- GS3.** convey and share technical information clearly using appropriate language
- GS4.** check and clarify task-related information
- GS5.** liaise with appropriate authorities using correct protocol
- GS6.** communicate with people in respectful form and manner in line with organizational protocol
- GS7.** undertake numerical operations, and calculations/ formulae
- GS8.**
 - identify and draw various basic, compound and solid shapes as per dimensions given Basic shapes: square, rectangle, triangle, circle
 - Compound shapes: involving squares, rectangles, triangles, circles, semi-circles, quadrants of a circle
 - Solid shapes: cube, rectangular prism, cylinder
- GS9.** use appropriate measuring techniques and units of measurement
- GS10.** use appropriate units and number systems to express degree of accuracy
- GS11.** use metric systems of measurement
- GS12.**
 - participate in on-the-job and other learning, training and development
 - interventions and assessments
- GS13.** clarify task related information with appropriate personnel or technical adviser
- GS14.** seek to improve and modify own work practices
- GS15.** maintain current knowledge of application standards, codes of practice and product/process developments
- GS16.** identify problems with work planning, procedures, output and behavior and their implications
- GS17.** prioritize and plan for problem solving
- GS18.** communicate problems appropriately to others
- GS19.** identify sources of information and support for problem solving
- GS20.** seek assistance and support from other sources to solve problems
- GS21.** identify effective resolution techniques
- GS22.** select and apply resolution techniques
- GS23.** seek evidence for problem resolution
- GS24.** plan, prioritize and sequence work operations as per job requirements
- GS25.** organize and analyze information relevant to work
- GS26.** basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time
- GS27.** undertake and express new ideas and initiatives to others
- GS28.** modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
- GS29.** participate in improvement procedures including process, quality and internal/external customer/supplier relationships
- GS30.** one competencies in new and different situations and contexts to achieve more



Qualification Pack

- GS31.** exercise restraint while expressing dissent and during conflict situations
- GS32.** avoid and manage distractions to be disciplined at work
- GS33.** manage own time for achieving better results
- GS34.** work in a team in order to achieve better results
- GS35.** identify and clarify work roles within a team
- GS36.** communicate and cooperate with others in the team for better results
- GS37.** seek assistance from fellow team members

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Working safely</i>	4.5	12	-	-
PC1. comply with health and safety, environmental and other relevant regulations and guidelines at work	0.75	2	-	-
PC2. adhere to procedures and guidelines for personal protective equipment (PPE) and other relevant safety regulations while performing turning operations	0.75	2	-	-
PC3. ensure work area is clean and safe from hazards	0.75	2	-	-
PC4. ensure that all tools, equipment, power tool cables, extension leads are in a safe and usable condition	0.75	2	-	-
PC5. ensure that machine guards are in place and are correctly fitted	0.75	2	-	-
PC6. read and understand safety instructions, warning signs on the machine	0.75	2	-	-
<i>Preparing for operating conventional turning machine</i>	9.75	13	-	-
PC7. ensure that all measuring equipment is within calibration due date	0.75	1	-	-
PC8. ensure availability of job specification from a valid source/Drawings	0.75	1	-	-
PC9. read and establish job requirements from the job specification document	0.75	1	-	-
PC10. ensure that the incoming components used are free from foreign objects, dirt or other contamination	0.75	1	-	-
PC11. prepare and maintain the work area as per procedure or operation specification	0.75	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. plan to carry out the required turning activities and the sequence of operations as per specifications	0.75	1	-	-
PC13. apply safe working practices and procedures at all times	0.75	1	-	-
PC14. obtain all the appropriate materials, cutting tools and measuring equipments required for the turning operation	0.75	1	-	-
PC15. confirm that the machine is ready for production	0.75	1	-	-
PC16. prepare for the turning activities by mounting, positioning and correctly setting a range of work holding devices and cutting tools	0.75	1	-	-
PC17. seek any necessary instruction/training on the operation of the machine, where required	0.75	1	-	-
PC18. hold components securely, without distortion	0.75	1	-	-
PC19. ensure that machine settings are adjusted as and when required to maintain the required accuracy	0.75	1	-	-
<i>Carrying out operations on conventional turning machine</i>	10.75	30	-	-
PC20. obtain the component drawings, specifications, job instructions required for the components to be machined	0.75	1	-	-
PC21. use and extract information from engineering drawings and related specifications	0.75	1	-	-
PC22. set and adjust the machine tool speeds and feeds to achieve the component specification	0.75	1	-	-
PC23. mount and set the required work holding devices, work piece and cutting tools	0.5	1	-	-
PC24. operate the machine tool controls safely and correctly, in line with operational procedures	0.5	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC25. control the machine in both hand and power modes for normal operations	0.5	1	-	-
PC26. Stop the machine in both normal and emergency situations correctly, and follow right procedure for restarting after an emergency	0.5	1	-	-
PC27. use lathes and the accessories that consists of saddle, capstan/turret head, compound slide, tailstock, taper turning attachments, profile attachments, fixed and travelling steadies	0.5	1	-	-
PC28. position and secure work holding devices to the machine spindle	0.5	1	-	-
PC29. perform turning operations using various equipments to produce components with various features	0.5	1	-	-
PC30. produce components as per given quality standards	0.5	2	-	-
PC31. plan and work to achieve given production targets	0.5	2	-	-
PC32. overcome the effects of backlash in machine slides and screws	0.5	2	-	-
PC33. perform the technique of trial cut for checking dimensional accuracy	0.5	2	-	-
PC34. apply roughing and finishing cuts, considering the effect on tool life, surface finish and dimensional accuracy	0.5	2	-	-
PC35. use cutting fluids for different materials	0.5	2	-	-
PC36. use range of measuring instruments to check critical parameters	0.5	2	-	-
PC37. clamp the work piece in a chuck/work holding device	0.5	2	-	-
PC38. perform the checks to be carried out on the components before removing them from the machine, and the equipment needed for this activity	0.5	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC39. ensure that the quality control procedures are used while operating the equipment	0.5	2	-	-
NOS Total	25	55	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0605
NOS Name	Operation on Lathe machine
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Tool Room Plastics, Tool Room Plastics
NSQF Level	3.5
Credits	2
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

CPC/CAP/N0606: To Perform Operation on Milling Machines

Description

This NOS covers performing milling operations on milling machine, to produce a range of components that combine a number of different features (eg. flat faces, parallel faces, faces that are flat and square to each other, angular faces, steps, slots, drilling and special forms). The candidate will be expected to work as per instructions given, taking personal responsibility for own actions and for the quality and accuracy of the work that they produce

Scope

The scope covers the following :

- This unit/task covers the following:
- Working safely
- Preparing for operating conventional milling machine
- Carrying out operations on conventional milling machine

Elements and Performance Criteria

Working safely

To be competent, the user/individual on the job must be able to:

- PC1.** comply with health and safety, environmental and other relevant regulations and guidelines at work
- PC2.** adhere to procedures and guidelines for personal protective equipment (PPE) and other relevant safety regulations while performing machining operations
- PC3.** work following laid down procedures and instructions
- PC4.** ensure work area is clean and safe from hazards
- PC5.** ensure that all tools, equipment, power tool cables, extension leads are in a safe and usable condition

Preparing for operating conventional milling machine

To be competent, the user/individual on the job must be able to:

- PC6.** ensure that all measuring instruments are within calibration due date
- PC7.** ensure that the components used are free from foreign objects, dirt or other contamination
- PC8.** ensure availability of job specification from a valid source
- PC9.**
 - read and establish job requirements from the job specification document
 - Job specifications documents: detailed component drawings; approved drawings as per national, international and organizational standards
- PC10.** prepare and maintain the work area as per procedure or operation specification
- PC11.** confirm that the machine is ready for production
- PC12.** seek any necessary instruction/training on the operation of the various milling machines, where appropriate
- PC13.** ensure that machine guards are in place and are correctly adjusted

Qualification Pack

- PC14.** identify different types of cutters used in horizontal and vertical milling machines
- PC15.** identify different parts of the vertical and horizontal milling machine
- PC16.** hold components securely, without distortion
- PC17.**
 - ensure that machine settings are adjusted as and when required to maintain
 - the required accuracy and quality standards Quality standards: components to be free from false tool cuts, burrs and sharp edges; dimensional tolerance 0.020 to 0.030 mm; flatness and squareness within 0.125mm; surface finish 1.6µm; angles within +/- 1 degree

Carrying out operations on conventional milling machine

To be competent, the user/individual on the job must be able to:

- PC18.** obtain the component drawings, specifications and job instructions required for the components to be machined
- PC19.** use and extract information from engineering drawings and related specifications to include symbols and conventions to appropriate ISO standards in relation to work undertaken
- PC20.** operate the machine controls in both hand and power modes
- PC21.** stop the machine in both normal and emergency situations, and use correct procedure for restarting after an emergency
- PC22.** use imperial and metric systems of measurement
- PC23.** perform various milling operations to produce various features on metal and non-metal components
- PC24.** produce components as per given quality standards
- PC25.** achieve given production targets
- PC26.** overcome the effects of backlash in machine slides and screws
- PC27.** apply roughing and finishing cuts considering the effect on tool life, surface finish and dimensional accuracy
- PC28.** apply cutting fluids with regard to a range of different materials
- PC29.** clamp the work piece securely and without distortion in a chuck/work holding device such as vice, V-block, clamp, angle plate, etc
- PC30.** ensure that the quality control procedures are used on the equipment
- PC31.** use range of equipment to check critical parameters

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** relevant health and safety requirements applicable in the work place
- KU3.** importance of working in clean and safe environment
- KU4.** own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
- KU5.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU6.** relevant people and their responsibilities within the work area
- KU7.** escalation matrix and procedures for reporting work and employment related issues

Qualification Pack

- KU8.** documentation and related procedures applicable in the context of employment and work
- KU9.** importance and purpose of documentation in context of employment and work
- KU10.** where personal protective equipment to be worn can be obtained
- KU11.** hazards associated with the milling operations and how they can be minimized
- KU12.** importance of keeping the work area clean and tidy
- KU13.** where to obtain the component drawings, specifications and/or job instructions required for them components to be machined
- KU14.** how to read and interpret first and third angle component drawings
- KU15.** how to extract information from engineering drawings or data and related specifications
- KU16.** how to use imperial and metric systems of measurement
- KU17.** main parts of conventional milling machines and the accessories that can be used
- KU18.** purpose and applications of milling
- KU19.** different types of milling cutters and their uses
- KU20.** various milling operations that can be performed, and the features produced on metal and non-metal components
- KU21.** processes of milling e.g. up milling, down milling, face milling, end milling, etc
- KU22.** effects of backlash in machine slides and screws, and how this can be overcome
- KU23.** effects of clamping the work piece in a chuck/work holding device, and how this can cause distortion in the finished components
- KU24.** production cost, machine hour rate, raw material cost, tool cost, coolant cost, overheads, cycle time, idle time, cost of machine idling, part rejection cost
- KU25.** selection of cutting tools, tool materials, chip breaker geometry, selecting cutting parameters from tool catalogues, selecting coolant
- KU26.** relationship between metal cutting results, tool Nose radius, speed and feed rate
- KU27.** how to recognize machining faults and how to identify when tools need re-sharpening
- KU28.** problems that can occur with the milling activities, and how these can be overcome
- KU29.** extent of their own authority and to whom they should report if they have problems that they cannot resolve
- KU30.** safe working practices and environmental regulations that must be observed
- KU31.** importance of reporting problems in a timely manner

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret information correctly from various job specification documents, manuals, health and safety instructions, memos, etc. applicable to the job in English and/or local language
- GS2.** fill up appropriate technical forms, process charts, activity logs as per organizational format in English and/or local language
- GS3.** convey and share technical information clearly using appropriate language
- GS4.** check and clarify task-related information

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- GS5.** liaise with appropriate authorities using correct protocol
- GS6.** communicate with people in respectful form and manner in line with organizational protocol
- GS7.** undertake numerical operations, and calculations/ formulae
- GS8.**
 - identify and draw various basic, compound and solid shapes as per dimensions given
 - Basic shapes: square, rectangle, triangle, circle
 - Compound shapes: involving squares, rectangles, triangles, circles, semi-circles, quadrants of a circle
 - Solid shapes: cube, rectangular prism, cylinder
- GS9.** use appropriate measuring techniques and units of measurement
- GS10.** use appropriate units and number systems to express degree of accuracy
- GS11.** use metric systems of measurement
- GS12.** participate in on-the-job and other learning, training and development interventions and assessments
- GS13.** clarify task related information with appropriate personnel or technical adviser
- GS14.** seek to improve and modify own work practices
- GS15.** maintain current knowledge of application standards, legislation, codes of practice and product/process developments
- GS16.** identify problems with work planning, procedures, output and behavior and their implications
- GS17.** prioritize and plan for problem solving
- GS18.** communicate problems appropriately to others
- GS19.** identify sources of information and support for problem solving
- GS20.** seek assistance and support from other sources to solve problems
- GS21.** identify effective resolution techniques
- GS22.** select and apply resolution techniques
- GS23.** seek evidence for problem resolution
- GS24.** plan, prioritize and sequence work operations as per job requirements
- GS25.** organize and analyze information relevant to work
- GS26.** basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time
- GS27.** undertake and express new ideas and initiatives to others
- GS28.** modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
- GS29.** participate in improvement procedures including process, quality and internal/external customer/supplier relationships
- GS30.** one competencies in new and different situations and contexts to achieve more
- GS31.** exercise restraint while expressing dissent and during conflict situations
- GS32.** avoid and manage distractions to be disciplined at work
- GS33.** manage own time for achieving better results
- GS34.** work in a team in order to achieve better results
- GS35.** identify and clarify work roles within a team
- GS36.** communicate and cooperate with others in the team for better results



Qualification Pack

GS37. seek assistance from fellow team members

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Working safely</i>	2.5	10	-	-
PC1. comply with health and safety, environmental and other relevant regulations and guidelines at work	0.5	2	-	-
PC2. adhere to procedures and guidelines for personal protective equipment (PPE) and other relevant safety regulations while performing machining operations	0.5	2	-	-
PC3. work following laid down procedures and instructions	0.5	2	-	-
PC4. ensure work area is clean and safe from hazards	0.5	2	-	-
PC5. ensure that all tools, equipment, power tool cables, extension leads are in a safe and usable condition	0.5	2	-	-
<i>Preparing for operating conventional milling machine</i>	7	24	-	-
PC6. ensure that all measuring instruments are within calibration due date	0.5	2	-	-
PC7. ensure that the components used are free from foreign objects, dirt or other contamination	0.5	2	-	-
PC8. ensure availability of job specification from a valid source	0.5	2	-	-
PC9. • read and establish job requirements from the job specification document • Job specifications documents: detailed component drawings; approved drawings as per national, international and organizational standards	0.5	2	-	-
PC10. prepare and maintain the work area as per procedure or operation specification	0.5	2	-	-
PC11. confirm that the machine is ready for production	0.5	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. seek any necessary instruction/training on the operation of the various milling machines, where appropriate	0.5	2	-	-
PC13. ensure that machine guards are in place and are correctly adjusted	1	2	-	-
PC14. identify different types of cutters used in horizontal and vertical milling machines	0.5	2	-	-
PC15. identify different parts of the vertical and horizontal milling machine	0.5	2	-	-
PC16. hold components securely, without distortion	0.5	2	-	-
PC17. - ensure that machine settings are adjusted as and when required to maintain - the required accuracy and quality standards Quality standards: components to be free from false tool cuts, burrs and sharp edges; dimensional tolerance 0.020 to 0.030 mm; flatness and squareness within 0.125mm; surface finish 1.6µm; angles within +/- 1 degree	1	2	-	-
<i>Carrying out operations on conventional milling machine</i>	10.5	26	-	-
PC18. obtain the component drawings, specifications and job instructions required for the components to be machined	1	2	-	-
PC19. use and extract information from engineering drawings and related specifications to include symbols and conventions to appropriate ISO standards in relation to work undertaken	1	2	-	-
PC20. operate the machine controls in both hand and power modes	1	2	-	-
PC21. stop the machine in both normal and emergency situations, and use correct procedure for restarting after an emergency	0.5	2	-	-
PC22. use imperial and metric systems of measurement	0.5	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. perform various milling operations to produce various features on metal and non-metal components	1	2	-	-
PC24. produce components as per given quality standards	0.5	2	-	-
PC25. achieve given production targets	0.5	1	-	-
PC26. overcome the effects of backlash in machine slides and screws	1	1	-	-
PC27. apply roughing and finishing cuts considering the effect on tool life, surface finish and dimensional accuracy	0.5	2	-	-
PC28. apply cutting fluids with regard to a range of different materials	1	2	-	-
PC29. clamp the work piece securely and without distortion in a chuck/work holding device such as vice, V-block, clamp, angle plate, etc	1	2	-	-
PC30. ensure that the quality control procedures are used on the equipment	0.5	2	-	-
PC31. use range of equipment to check critical parameters	0.5	2	-	-
NOS Total	20	60	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0606
NOS Name	To Perform Operation on Milling Machines
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Tool Room Plastics, Tool Room Plastics
NSQF Level	3.5
Credits	2
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

CPC/CAP/N0607: To Perform Operation on Grinding machines

Description

This unit covers grinding of various components required in the manufacturing sector using different types of Grinding machines. This will involve carrying out the grinding operations like horizontal, vertical surfaces, cylindrical, angular, taper surfaces, and grinding of single point and multipoint cutters in accordance with approved procedures using different types of grinding machines i.e. surface grinder, cylindrical grinder, pedestal grinder, bench grinder, tool and cutter grinder grinding machines). The candidate will be expected to work as per instructions given, taking personal responsibility for own actions and for the quality and accuracy of the work that they produce

Scope

The scope covers the following :

- This unit/task covers the following:
- Working safely
- Operating Grinding Machine
- Handling of unresolved problems
- Processing Compliances

Elements and Performance Criteria

Working safely

To be competent, the user/individual on the job must be able to:

- PC1.** comply with health and safety, environmental and other relevant regulations and guidelines at work
- PC2.** adhere to procedures and guidelines for personal protective equipment (PPE) and other relevant safety rules and regulations
- PC3.** work following laid down procedures and instructions
- PC4.** ensure work area is clean and safe from hazards
- PC5.** ensure that all tools, equipment, power tool cables, extension leads are in a safe and usable condition

Operating Grinding Machine

To be competent, the user/individual on the job must be able to:

- PC6.** check that all measuring instruments are within calibration due date
- PC7.** obtain and prepare the appropriate materials, tools and measuring instruments
- PC8.** mount the work-piece safely and securely, in line with instructions
- PC9.** set and adjust the grinding machine speed and feed, in line with instructions
- PC10.** use the grinding machine controls safely and correctly, in line with operational procedures
- PC11.** check that the finished components meet the drawing dimensions as required
- PC12.** report any difficulties or problems that may arise during grinding activities, and carry out suitable actions

Qualification Pack

- PC13.** shut down the machine to a safe condition on completion of the grinding activities
- PC14.** prepare grinding wheels through various methods
- PC15.** Grind components to produce various features
- PC16.** check the quality of output, using measuring equipment appropriate to the aspects being checked and the tolerances to be achieved
- PC17.** check the machined component for accuracy in dimensions, parallelism and surface texture as per job specifications

Handling of unresolved problems

To be competent, the user/individual on the job must be able to:

- PC18.** refer the problem to a competent internal specialist if it cannot be resolved
- PC19.** obtain help or advice from specialist if the problem is outside his/her area of competence or experience

Processing Compliances

To be competent, the user/individual on the job must be able to:

- PC20.** comply with relevant standards, policies and procedures

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** relevant health and safety requirements applicable in the work place
- KU3.** importance of working in clean and safe environment
- KU4.** own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
- KU5.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU6.** relevant people and their responsibilities within the work area
- KU7.** procedures for reporting work and employment related issues
- KU8.** documentation and related procedures applicable related to employment and work
- KU9.** importance and purpose of documentation related to employment and work
- KU10.** safety mechanisms on the machine, and the procedure for checking that they function correctly
- KU11.** correct operation of the grinding machine controls in both manual and power modes how to stop the machine in both normal and emergency situations, and the procedure for restarting after an emergency
- KU12.** importance of keeping the work area clean and tidy (eg. cleaning the machine, disposal of waste, ensuring any spilt cutting fluids are correctly dealt with)
- KU13.** how to use and extract information from engineering drawings and related specifications (to include ISO standard symbols and abbreviations, imperial and metric systems of measurement, work-piece reference points and system of tolerance)
- KU14.** how to mount the work-piece in the work-holding devices

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- KU15.** effects of clamping the work-piece in a chuck/work holding device, and how this can cause damage or distortion in the finished components
- KU16.** how to check that the grinding wheels are in a safe and serviceable condition (eg. free from damage, cracks, correctly balanced)
- KU17.** need for 'trueing up' and dressing of wheels to prevent glazing and burning of the work-piece, and methods of forming the wheels to the required profile (eg. use of pantograph, diamond dressing units)
- KU18.** effects of backlash in machine slides and screws, and how this can be overcome
- KU19.** techniques of taking trial cuts and checking dimensional accuracy
- KU20.** application of roughing and finishing cuts, and the effect on tool life, surface finish and dimensional accuracy
- KU21.** types of grinding wheels, cutting feeds and speeds to be used, and the depth of cut that can be taken
- KU22.** application of cutting fluids with regard to a range of different materials, and why some materials do not require cutting fluids to be used
- KU23.** how to recognize grinding faults, and how to identify when grinding wheels need dressing
- KU24.** checks to be carried out on the components before removing them from the machine (eg. have all operations been completed, dimensional checks, surface finish checks)
- KU25.** problems that can occur with the grinding activities and how to address them
- KU26.** importance of leaving the machine in a safe condition on completion of activities
- KU27.** safe working practices and procedures to be followed when preparing and using grinding machines
- KU28.** hazards associated with the grinding operations and how they can be minimized
- KU29.** personal protective equipment (PPE) to be worn for the grinding activities and personal safety measures taken

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret information correctly from various job specification documents, manuals, health and safety instructions, etc. applicable to the job
- GS2.** fill up appropriate technical forms, process charts, activity logs as per organizational format
- GS3.** convey and share technical information clearly using appropriate language
- GS4.** check and clarify task-related information
- GS5.** liaise with appropriate authorities using correct protocol
- GS6.** communicate with people in respectful form and manner in line with organizational protocol
- GS7.** undertake numerical operations, and calculations/ formulae
- GS8.** identify and draw various basic, compound and solid shapes as per dimensions given
- GS9.** use appropriate measuring techniques and units of measurement
- GS10.** use appropriate units and number systems to express degree of accuracy
- GS11.** use metric systems of measurement

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- GS12.** participate in on-the-job and other learning, training and development interventions and assessments
- GS13.** clarify task related information with appropriate personnel or technical adviser
- GS14.** seek to improve and modify own work practices
- GS15.** maintain current knowledge of application standards, codes of practice and product/process developments
- GS16.** identify problems with work planning, procedures, output and behavior and their implications
- GS17.** prioritize and plan for problem solving
- GS18.** communicate problems appropriately to others
- GS19.** identify sources of information and support for problem solving
- GS20.** seek assistance and support from other sources to solve problems
- GS21.** identify effective resolution techniques
- GS22.** select and apply resolution techniques
- GS23.** seek evidence for problem resolution
- GS24.** plan, prioritize and sequence work operations as per job requirements
- GS25.** organize and analyze information relevant to work
- GS26.** basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time
- GS27.** undertake and express new ideas and initiatives to others
- GS28.** modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
- GS29.** participate in improvement procedures including process, quality and customer relationships
- GS30.** one competencies in new and different situations and contexts to achieve more
- GS31.** exercise restraint while expressing dissent and during conflict situations
- GS32.** avoid and manage distractions to be disciplined at work
- GS33.** manage own time for achieving better results
- GS34.** work in a team in order to achieve better results
- GS35.** identify and clarify work roles within a team
- GS36.** communicate and cooperate with others in the team for better results
- GS37.** seek assistance from fellow team members

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Working safely</i>	5	7.5	-	-
PC1. comply with health and safety, environmental and other relevant regulations and guidelines at work	1	1.5	-	-
PC2. adhere to procedures and guidelines for personal protective equipment (PPE) and other relevant safety rules and regulations	1	1.5	-	-
PC3. work following laid down procedures and instructions	1	1.5	-	-
PC4. ensure work area is clean and safe from hazards	1	1.5	-	-
PC5. ensure that all tools, equipment, power tool cables, extension leads are in a safe and usable condition	1	1.5	-	-
<i>Operating Grinding Machine</i>	12	18	-	-
PC6. check that all measuring instruments are within calibration due date	1	1.5	-	-
PC7. obtain and prepare the appropriate materials, tools and measuring instruments	1	1.5	-	-
PC8. mount the work-piece safely and securely, in line with instructions	1	1.5	-	-
PC9. set and adjust the grinding machine speed and feed, in line with instructions	1	1.5	-	-
PC10. use the grinding machine controls safely and correctly, in line with operational procedures	1	1.5	-	-
PC11. check that the finished components meet the drawing dimensions as required	1	1.5	-	-
PC12. report any difficulties or problems that may arise during grinding activities, and carry out suitable actions	1	1.5	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. shut down the machine to a safe condition on completion of the grinding activities	1	1.5	-	-
PC14. prepare grinding wheels through various methods	1	1.5	-	-
PC15. Grind components to produce various features	1	1.5	-	-
PC16. check the quality of output, using measuring equipment appropriate to the aspects being checked and the tolerances to be achieved	1	1.5	-	-
PC17. check the machined component for accuracy in dimensions, parallelism and surface texture as per job specifications	1	1.5	-	-
<i>Handling of unresolved problems</i>	2	3	-	-
PC18. refer the problem to a competent internal specialist if it cannot be resolved	1	1.5	-	-
PC19. obtain help or advice from specialist if the problem is outside his/her area of competence or experience	1	1.5	-	-
<i>Processing Compliances</i>	1	1.5	-	-
PC20. comply with relevant standards, policies and procedures	1	1.5	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0607
NOS Name	To Perform Operation on Grinding machines
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Tool Room Plastics, Tool Room Plastics
NSQF Level	3.5
Credits	2
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

CPC/CAP/N0608: To Preform Basic programming and operation on CNC Machines

Description

This NOS covers making programs for and proving out of parts on Computer Numerically Controlled (CNC) lathes, milling and EDM machines. Programming can be done manually or using CAM software. The program is transferred to the machine's controller by entering it at the console, transmitting it through a wired link, or copying it through a data storage device like a flash card. The candidate will be expected to perform safe operations with a minimum of supervision, taking personal responsibility for one own actions and for the quality and accuracy of the work produced.

Scope

The scope covers the following :

- This task covers the following:
- Working safely
- Preparing for programming CNC Lathes, Milling and EDM machine for production
- Carrying out programming for CNC machine
- Test run and proving the program on CNC Machine

Elements and Performance Criteria

Working safely

To be competent, the user/individual on the job must be able to:

- PC1.** comply with health and safety, environmental and other relevant regulations and guidelines at work
- PC2.** adhere to procedures and guidelines for personal protective equipment (PPE) and other relevant safety regulations while programming CNC machines
- PC3.** work following laid down procedures and instructions
- PC4.** ensure that machine guards are in place and are correctly adjusted
- PC5.** read and understand safety instructions, warning signs on the machine
- PC6.** ensure work area is clean and safe from hazards
- PC7.** ensure that all the cutting tools, measuring instruments, cables, extension leads are in a safe and usable condition

Preparing for programming CNC machine for production

To be competent, the user/individual on the job must be able to:

- PC8.** obtain job specification from a valid and approved source
- PC9.** read and establish job requirements from the job specification document accurately
- PC10.** follow job instructions, assembly drawings and laid down procedures at all times
- PC11.** report and rectify incorrect and inconsistent information in job specification documents as per organization procedures
- PC12.** use and extract information from reference charts, tables, graphs and standards

Qualification Pack

- PC13.** prepare the work area as per procedure or operational specification
- PC14.** conduct a preliminary check of the readiness of the program so that the CNC machine operates correctly
- PC15.** determine what operational objectives and targets need to be achieved and how best the machine needs to be programmed to achieve this
- PC16.** extract and use information from engineering drawings and related specifications in relation to work undertaken
- PC17.** identify tool requirements from tooling layout and assess their suitability
- PC18.** identify suitable work holding or fixturing device as per the job requirement
- PC19.** ensure the correct and latest part-program is uploaded onto the CNC system
- PC20.** • Use Electric Discharge Machining to hole out blind spots and also to create
• hole in the die formation plate/ work piece
- PC21.** • Setup the electrodes of the EDM machine and measure the distance between
• the electrodes as mentioned in the Work Instructions
- PC22.** Ensure that the correct current and voltage are selected for the EDM process
- PC23.** • Ensure that the work piece/ metal piece is carefully loaded on the EDM
• machine surface tables/ work platform using manual/ automatic tools
- PC24.** • Ensure that there is uniform flow of dielectric liquid i.e. flushing of the
• dielectric liquid to remove any debris which would have collected during the
• EDM process
- PC25.** • Ensure that the machine operations are regularly monitored to detect any
• malfunctions in machine operations or any out of tolerance machining
- PC26.** • Ensure that the electrode properties like surface, dimensions, metallurgical
• properties are periodically checked as per the checklist provided
- PC27.** • Ensure that the electrodes are changed in case there is a deviation from the
• specifications

Carrying out programming for CNC machine

To be competent, the user/individual on the job must be able to:

- PC28.** Prepare the CNC program with commands for tool motions, spindle motions, miscellaneous functions and tool change, in syntax corresponding to the machine and control system on which the component will be machined
- PC29.** various ways to make CNC program are by writing it on paper or in a computer's text editor, or using CAM software or controllers on machine
- PC30.** ensure that the part program is efficient and results in minimal cycle time, with optimal cutting parameters and no unnecessary tool motions
- PC31.** use subprograms and canned cycles, to reduce program size and input time and avoid memory overflow on the machine
- PC32.** transfer the program to the machine by entering it at the console or transmitting it through a wired link or through a data transfer device
- PC33.** follow the correct procedures for calling up the program and dealing with any error messages or faults
- PC34.** handle the typical problems that can occur with the programming, loading and editing activities effectively using approved procedures

Qualification Pack

PC35. save the proven program in the appropriate storage medium – paper, computer hard disk, etc. - and location

PC36. complete relevant documentation as per procedure

PC37. leave the work area in a safe and tidy condition on completion of the activities

Test and prove the program on the CNC Machine

To be competent, the user/individual on the job must be able to:

PC38. obtain appropriate equipment or tools needed as per job requirements

PC39. ensure that all measuring equipment is calibrated and approved for usage

PC40. ensure that the tools and fixtures are in usable condition(eg. free from breakage, damage, calibration, etc.)

PC41. pre-set the tooling appropriately using setting jigs/fixtures

PC42. seek any necessary instruction/training on the operation of the machine where required

PC43. mount tools in the correct positions in the tool turret or magazine

PC44. check that the tools have been mounted in positions corresponding to tool numbers in the part program

PC45. measure tool and work offset data - X and Z offsets for lathes; work offsets, length offsets and tool radius for machining centers

PC46. ensure that the component is free of burrs, chips or other material adhering to its butting surfaces

PC47. mount the part on machine firmly in the specified work holding devices, with the appropriate clamping forces

PC48. enter work offset and tool data on the machine – X and Z offsets, tool orientation and Nose radius for lathes; length offsets and tool radius for machining centers

PC49. ensure that tool data has been entered in offset number corresponding to the tool offset numbers in the part program

PC50. deal with error messages and faults on the program or equipment

PC51. • cut a trial part using single block run, dry run and feed and speed override
• controls

PC52. edit the program and adjust tool and wear offsets to correct any dimensional errors on the part

PC53. ensure that the trial part conforms to drawing specifications in terms of dimensions, surface finishes and geometrical parameters like concentricity, parallelism, run out, etc.

PC54. hand-over the machine to the machine operator for machining the batch of parts, along with relevant instructions and documentation on periodic inspection of components, change of worn out tools

PC55. correct the tool wear offsets whenever required, based on the results of the period inspection

PC56. change worn out tools and indexable inserts whenever required

PC57. after every change of a worn out tool or insert, cut a trial part and correct any dimensional inaccuracies by adjusting the tool offsets or wear offsets

PC58. return worn out cutting tools, work holding device / fixtures / instruments / drawings to store

PC59. ensure that there is no damage to the tool/fixture while doing the prove-out

PC60. shut down the equipment to a safe condition on conclusion of the activities

Qualification Pack

- PC61.** deal promptly and effectively with problems within span of responsibility and control and report those that cannot be solved

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** relevant health and safety requirements applicable in the work place
- KU3.** importance of working in clean and safe environment
- KU4.** own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
- KU5.**
 - reporting structure, inter-dependent functions, lines and procedures in the
 - work area
- KU6.** relevant people and their responsibilities within the work area
- KU7.** procedures for reporting work and employment related issues
- KU8.** documentation and related procedures applicable related to employment and work
- KU9.** importance and purpose of documentation in context of employment and work
- KU10.** specific safe working practices, CNC programming procedures and environmental regulations that must be observed
- KU11.** hazards associated with carrying out the machining operations on a CNC machine and how can they be minimized
- KU12.** personal protective equipment to be used during the machining activities on a CNC machine and where can it be obtained
- KU13.** safety mechanism on the machine and how to check if they are functioning properly
- KU14.** types and sources of appropriate job specifications
- KU15.** common terminology used in CNC programming
- KU16.** selection of strategies based on material and fixturing, holding and clamping force
- KU17.** the factors which will determine selection and use of tungsten carbide and tips
- KU18.** importance of tool selection based on material, finish required and tolerances achieved
- KU19.** importance of cutter engagement and exit
- KU20.** factors affecting tool life
- KU21.** importance and effect of the depth of cut, RPM and feed
- KU22.** how to read and interpret first and third angle component drawings
- KU23.** how to extract information from engineering drawings or data and related specifications
- KU24.** how to use the function keys and user interface of the machine control system
- KU25.** determination and entry of work and tool offsets, tool wear data
- KU26.** main features and working parts of the CNC machine, and the accessories that can be used
- KU27.** importance of following specified machining sequences and procedures
- KU28.** importance of ensuring suitability of workpieces/materials and consumables for the specified job and related procedures

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- KU29.** importance and procedures to ensure that tools and equipment are in a safe and usable condition
- KU30.** various CNC operations that can be performed, and the methods and equipment used
- KU31.** methods of setting the work-holding devices, and the tools and equipment that can be used
- KU32.** various tool holding devices that are used, and the methods of correctly mounting and securing the cutting tools to the tool holders
- KU33.** how to set the machine controller in the program and editing mode, and enter or download the prepared program
- KU34.** automatic tool changers, pallet changers, rotary tables and part auto loaders used
- KU35.** how to position and identify the tools in relationship to the operating program
- KU36.** function of error messages, and appropriate subsequent action
- KU37.** importance of proving the program, how to do it and selecting the correct proving tools in CNC Milling, CNC Lathe
- KU38.** Proper selection of Copper, graphite electrode in CNC-EDM, spark-gap
- KU39.** need for storing program tapes and disks safely and correctly, away from contaminants and electromagnetic sources
- KU40.** quality control procedures that are used, inspection checks to be carried out, and the equipment that will need to be used
- KU41.** importance to report problems in a timely manner
- KU42.** importance of writing programs that are easily editable or correctable by the next person
- KU43.** methods of checking quality of the shaped components against the required quality standards
- KU44.** production cost, machine hour rate, raw material cost, tool cost, coolant cost, overheads, cycle time, idle time, cost of machine idling, part rejection cost
- KU45.** selection of cutting tools, tool materials, chip breaker geometry, selecting cutting parameters from tool catalogues, selecting coolant
- KU46.** relationship between surface finish, tool Nose radius and feed rate
- KU47.** impact of depth of cut on chatter, surface finish
- KU48.** range of materials used in common engineering applications
- KU49.** how to identify materials by their physical properties

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret information correctly from various job specification documents, manuals, health and safety instructions, memos, etc. applicable to the job in English and/or local language
- GS2.** fill up appropriate technical forms, process charts, activity logs as per organizational format in English and/or local language
- GS3.** convey and share technical information clearly using appropriate language
- GS4.** check and clarify task-related information
- GS5.** liaise with appropriate authorities using correct protocol

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- GS6.** communicate with people in respectful form and manner in line with organizational protocol
- GS7.** undertake numerical computations and calculation
- GS8.**
- identify and draw various basic, compound and solid shapes as per dimensions given
 - Basic shapes: square, rectangle, triangle, circle, quadrilaterals
 - Compound shapes: involving squares, rectangles, triangles, circles, semi-circles, quadrants of a circle
 - Solid shapes: cube, rectangular prism, cylinder
- GS9.** use appropriate measuring techniques and units of measurement
- GS10.** use appropriate units and number systems to express degree of accuracy
- GS11.** interpret and express tolerance in terms of limits on dimensions
- GS12.** calculation of the value of angles in a triangle
- GS13.** participate in on-the-job and other learning, training and development interventions and assessments
- GS14.** clarify task related information with appropriate personnel or technical adviser
- GS15.** seek to improve and modify own work practices
- GS16.** maintain current knowledge of application standards, legislation, codes of practice and product/process developments
- GS17.** identify problems with work planning, procedures, output and behavior and their implications
- GS18.** prioritize and plan for problem solving
- GS19.** communicate problems appropriately to others
- GS20.** identify sources of information and support for problem solving
- GS21.** seek assistance and support from other sources to solve problems
- GS22.** identify effective resolution techniques
- GS23.** select and apply resolution techniques
- GS24.** seek evidence for problem resolution
- GS25.** plan, prioritize and sequence work operations as per job requirements
- GS26.** organize and analyze information relevant to work
- GS27.** basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time
- GS28.** undertake and express new ideas and initiatives to others
- GS29.** modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
- GS30.** participate in improvement procedures including process, quality and internal/external customer/supplier relationships
- GS31.** one competencies in new and different situations and contexts to achieve more
- GS32.** exercise restraint while expressing dissent and during conflict situations
- GS33.** avoid and manage distractions to be disciplined at work
- GS34.** manage own time for achieving better results
- GS35.** work in a team in order to achieve better results
- GS36.** identify and clarify work roles within a team
- GS37.** communicate and cooperate with others in the team for better results



Qualification Pack

GS38. seek assistance from fellow team members

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Working safely</i>	5.25	7	-	-
PC1. comply with health and safety, environmental and other relevant regulations and guidelines at work	0.75	1	-	-
PC2. adhere to procedures and guidelines for personal protective equipment (PPE) and other relevant safety regulations while programming CNC machines	0.75	1	-	-
PC3. work following laid down procedures and instructions	0.75	1	-	-
PC4. ensure that machine guards are in place and are correctly adjusted	0.75	1	-	-
PC5. read and understand safety instructions, warning signs on the machine	0.75	1	-	-
PC6. ensure work area is clean and safe from hazards	0.75	1	-	-
PC7. ensure that all the cutting tools, measuring instruments, cables, extension leads are in a safe and usable condition	0.75	1	-	-
<i>Preparing for programming CNC machine for production</i>	15	35	-	-
PC8. obtain job specification from a valid and approved source	0.75	1	-	-
PC9. read and establish job requirements from the job specification document accurately	0.75	1	-	-
PC10. follow job instructions, assembly drawings and laid down procedures at all times	0.75	1	-	-
PC11. report and rectify incorrect and inconsistent information in job specification documents as per organization procedures	0.75	1	-	-
PC12. use and extract information from reference charts, tables, graphs and standards	0.75	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. prepare the work area as per procedure or operational specification	0.75	2	-	-
PC14. conduct a preliminary check of the readiness of the program so that the CNC machine operates correctly	0.75	2	-	-
PC15. determine what operational objectives and targets need to be achieved and how best the machine needs to be programmed to achieve this	0.75	2	-	-
PC16. extract and use information from engineering drawings and related specifications in relation to work undertaken	0.75	2	-	-
PC17. identify tool requirements from tooling layout and assess their suitability	0.75	2	-	-
PC18. identify suitable work holding or fixturing device as per the job requirement	0.75	2	-	-
PC19. ensure the correct and latest part-program is uploaded onto the CNC system	0.75	2	-	-
PC20. • Use Electric Discharge Machining to hole out blind spots and also to create • hole in the die formation plate/ work piece	0.75	2	-	-
PC21. • Setup the electrodes of the EDM machine and measure the distance between • the electrodes as mentioned in the Work Instructions	0.75	2	-	-
PC22. Ensure that the correct current and voltage are selected for the EDM process	0.75	2	-	-
PC23. • Ensure that the work piece/ metal piece is carefully loaded on the EDM • machine surface tables/ work platform using manual/ automatic tools	0.75	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. - Ensure that there is uniform flow of dielectric liquid i.e. flushing of the - dielectric liquid to remove any debris which would have collected during the - EDM process	0.75	2	-	-
PC25. - Ensure that the machine operations are regularly monitored to detect any - malfunctions in machine operations or any out of tolerance machining	0.75	2	-	-
PC26. - Ensure that the electrode properties like surface, dimensions, metallurgical - properties are periodically checked as per the checklist provided	0.75	2	-	-
PC27. - Ensure that the electrodes are changed in case there is a deviation from the - specifications	0.75	2	-	-
<i>Carrying out programming for CNC machine</i>	7.5	14	-	-
PC28. Prepare the CNC program with commands for tool motions, spindle motions, miscellaneous functions and tool change, in syntax corresponding to the machine and control system on which the component will be machined	0.75	2	-	-
PC29. various ways to make CNC program are by writing it on paper or in a computer's text editor, or using CAM software or controllers on machine	0.75	2	-	-
PC30. ensure that the part program is efficient and results in minimal cycle time, with optimal cutting parameters and no unnecessary tool motions	0.75	2	-	-
PC31. use subprograms and canned cycles, to reduce program size and input time and avoid memory overflow on the machine	0.75	2	-	-
PC32. transfer the program to the machine by entering it at the console or transmitting it through a wired link or through a data transfer device	0.75	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC33. follow the correct procedures for calling up the program and dealing with any error messages or faults	0.75	1	-	-
PC34. handle the typical problems that can occur with the programming, loading and editing activities effectively using approved procedures	0.75	1	-	-
PC35. save the proven program in the appropriate storage medium – paper, computer hard disk, etc. - and location	0.75	1	-	-
PC36. complete relevant documentation as per procedure	0.75	1	-	-
PC37. leave the work area in a safe and tidy condition on completion of the activities	0.75	1	-	-
<i>Test and prove the program on the CNC Machine</i>	12.25	24	-	-
PC38. obtain appropriate equipment or tools needed as per job requirements	0.75	1	-	-
PC39. ensure that all measuring equipment is calibrated and approved for usage	0.5	1	-	-
PC40. ensure that the tools and fixtures are in usable condition(eg. free from breakage, damage, calibration, etc.)	0.5	1	-	-
PC41. pre-set the tooling appropriately using setting jigs/fixtures	0.5	1	-	-
PC42. seek any necessary instruction/training on the operation of the machine where required	0.5	1	-	-
PC43. mount tools in the correct positions in the tool turret or magazine	0.5	1	-	-
PC44. check that the tools have been mounted in positions corresponding to tool numbers in the part program	0.5	1	-	-
PC45. measure tool and work offset data - X and Z offsets for lathes; work offsets, length offsets and tool radius for machining centers	0.5	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC46. ensure that the component is free of burrs, chips or other material adhering to its butting surfaces	0.5	1	-	-
PC47. mount the part on machine firmly in the specified work holding devices, with the appropriate clamping forces	0.5	1	-	-
PC48. enter work offset and tool data on the machine – X and Z offsets, tool orientation and Nose radius for lathes; length offsets and tool radius for machining centers	0.5	1	-	-
PC49. ensure that tool data has been entered in offset number corresponding to the tool offset numbers in the part program	0.5	1	-	-
PC50. deal with error messages and faults on the program or equipment	0.5	1	-	-
PC51. • cut a trial part using single block run, dry run and feed and speed override • controls	0.5	1	-	-
PC52. edit the program and adjust tool and wear offsets to correct any dimensional errors on the part	0.5	1	-	-
PC53. ensure that the trial part conforms to drawing specifications in terms of dimensions, surface finishes and geometrical parameters like concentricity, parallelism, run out, etc.	0.5	1	-	-
PC54. hand-over the machine to the machine operator for machining the batch of parts, along with relevant instructions and documentation on periodic inspection of components, change of worn out tools	0.5	1	-	-
PC55. correct the tool wear offsets whenever required, based on the results of the period inspection	0.5	1	-	-
PC56. change worn out tools and indexable inserts whenever required	0.5	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC57. after every change of a worn out tool or insert, cut a trial part and correct any dimensional inaccuracies by adjusting the tool offsets or wear offsets	0.5	1	-	-
PC58. return worn out cutting tools, work holding device / fixtures / instruments / drawings to store	0.5	1	-	-
PC59. ensure that there is no damage to the tool/fixture while doing the prove-out	0.5	1	-	-
PC60. shut down the equipment to a safe condition on conclusion of the activities	0.5	1	-	-
PC61. deal promptly and effectively with problems within span of responsibility and control and report those that cannot be solved	0.5	1	-	-
NOS Total	40	80	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0608
NOS Name	To Preform Basic programming and operation on CNC Machines
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Tool Room Plastics, Tool Room Plastics
NSQF Level	3.5
Credits	2
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

CPC/CAP/N0403: Effective working with others.

Description

This unit covers basic etiquette and competencies that an individual is required to possess and demonstrate in their behavior and interactions with others at the workplace. This covers areas such as communication, discipline, handling conflict and grievances.

Scope

The scope covers the following :

- This unit/task covers the following:
- Effective Working with others

Elements and Performance Criteria

Effective Working with others

To be competent, the user/individual on the job must be able to:

- PC1.** • display appropriate communication etiquette while working
 - Communication etiquette: use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism, use appropriate titles and terms of respect; do not use abusive language
- PC2.** display active listening skills while interacting with others at work
- PC3.** demonstrate responsible & disciplined behaviors at workplace
- PC4.** accurately receive information & instructions from supervisor & fellow workers, getting clarification where required
- PC5.** accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt
- PC6.** display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible
- PC7.** consult with and assist others to maximize effectiveness and efficiency in carrying out tasks
- PC8.** escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** policies and procedures followed in the company for working with others in an organizational set-up
- KU2.** grievance/ conflict handling mechanism of the company
- KU3.** relevant people and their responsibilities within the work area
- KU4.** importance of effective communication in the workplace



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- KU5.** importance of teamwork in organizational and individual success
- KU6.** barriers to effective communication
- KU7.** importance of avoiding casual expletives and unpleasant terms while communicating professional circles
- KU8.** various categories of people that one is required to communicate and co- ordinate within the organization
- KU9.** importance of discipline for professional success
- KU10.** importance of ethics for professional success
- KU11.** what constitutes disciplined behavior for a working professional
- KU12.** common reasons for interpersonal conflict
- KU13.** importance and ways of managing interpersonal conflict effectively
- KU14.** importance of developing effective working relationships for professional success
- KU15.** expressing and addressing grievances appropriately and effectively

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Effective Working with others</i>	10	10	-	-
PC1. - display appropriate communication etiquette while working - Communication etiquette: use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism, use appropriate titles and terms of respect; do not use abusive language	1	1	-	-
PC2. display active listening skills while interacting with others at work	1	1	-	-
PC3. demonstrate responsible & disciplined behaviors at workplace	1	1	-	-
PC4. accurately receive information & instructions from supervisor & fellow workers, getting clarification where required	1	1	-	-
PC5. accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt	2	2	-	-
PC6. display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible	2	2	-	-
PC7. consult with and assist others to maximize effectiveness and efficiency in carrying out tasks	1	1	-	-
PC8. escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict	1	1	-	-
NOS Total	10	10	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0403
NOS Name	Effective working with others.
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Generic, Generic
NSQF Level	3.5
Credits	1
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

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PC6. work with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

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- KU8.** different types of disabilities and appropriate communication and behaviour towards PwD
- KU9.** different types of financial products and services
- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment System Overview:

- Batches are assigned to Training Assessment Wing (TAW), CIPET HO for planning of assessment
- Training Centres, requesting TAW for Assessment and Certification of Trainees
- TAW identifies suitable assessor and nominates the assessor to the respective Training Centre
- TAW monitors the assessment process
- Training Centres are maintaining the records

2. Testing Environment:

- Check the Assessment location, date and time
- If the batch size is more than 30, then there should be 02 Assessors in a day (or) 01 assessor in 2 days

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- Check that the allotted time to the candidates to complete the Theory & Practical Assessment

3. Assessment Quality Assurance levels/Framework:

- Question bank / Question Paper is prepared by the Subject Matter Experts (SME) / Assessor
- Questions are mapped to the specified assessment criteria
- Certified Assessor & Trainer will be engaged in the process

4. Types of evidence or evidence-gathering protocol:

5. Method of verification or validation:

- Date / Time recorded for the reporting of the assessor from assessment location
- Assessment batch - Group Photo of Trainees along with Assessor
- Surprise visit to the assessment location
- Virtual meet with the Assessor / Trainees

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CPC/CAP/N0401.Basic health and safety practices at the workplace, 5S	10	30	-	-	40	10
CPC/CAP/N0601.To Understanding the basic concepts, design, drawings and planning for machining components, making tools & dies and coordinating with others	10	20	0	0	30	10

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CPC/CAP/N0602.To Perform fitting operations on machining components using hand tools	15	25	0	0	40	10
CPC/CAP/N0603.To Operation on Drilling	20	40	0	0	60	10
CPC/CAP/N0604.To Perform Operation on Shaping Machine	20	40	0	0	60	10
CPC/CAP/N0605.Operation on Lathe machine	25	55	0	0	80	10
CPC/CAP/N0606.To Perform Operation on Milling Machines	20	60	0	0	80	10
CPC/CAP/N0607.To Perform Operation on Grinding machines	20	30	0	0	50	10
CPC/CAP/N0608.To Preform Basic programming and operation on CNC Machines	40	80	0	0	120	10
CPC/CAP/N0403.Effective working with others.	10	10	0	0	20	5
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	5
Total	210	420	-	-	630	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.